

Software Requirements Specification (SRS)

AI-Powered Anomaly Detection System for MPLADS Scheme

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Document Control

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SECTION 1: DOCUMENT OVERVIEW

1.1 Purpose

This SRS defines the complete, machine-readable, AI-executable specification for the Minimum Viable Product (MVP) of "**MPLADS Sentinel**" — an AI-powered monitoring and analytics platform designed to detect anomalies, fraud, and inefficiencies in the Members of Parliament Local Area Development Scheme (MPLADS) implementation. Every requirement in this document is written as an unambiguous, testable instruction that an AI agentic development system can interpret, implement, and verify without human clarification.

1.2 Scope

1.2.1 In-Scope (MVP Boundary)

YAML

MVP_SCOPE:

DATA_INGESTION:

- Ingest MPLADS data from public dashboard CSV/API exports
- Parse and normalize sanction, expenditure, work progress, and payment records
- Store in structured relational + vector database schema

ANOMALY_DETECTION:

- Statistical anomaly detection on cost estimates vs actual expenditure
- Duplicate work detection using NLP similarity on work descriptions
- Delayed project identification based on timeline deviation thresholds
- Cost overrun flagging using z-score and IQR methods
- Fund utilization rate analysis per constituency per fiscal year

FRAUD_RISK_SCORING:

- Composite risk score per work item (0-100 scale)
- Rule-based + ML-based hybrid scoring model
- Classification of risk into LOW / MEDIUM / HIGH / CRITICAL tiers

DASHBOARD:

- Web-based responsive dashboard with role-based views
- National overview heatmap
- Constituency-level drill-down
- Alert feed with severity indicators
- Trend analysis charts (time-series)

ALERTING:

- Automated alert generation on anomaly detection
- Alert categorization: COST_OVERRUN | DUPLICATE_WORK | DELAYED_PROJECT | FUND_MISUTILIZATION | PATTERN_ANOMALY
- Alert log with audit trail

REPORTING:

- Exportable PDF/CSV reports per constituency
- Summary analytics for ministry-level review

1.2.2 Out-of-Scope (Post-MVP)

YAML

OUT_OF_SCOPE:

- Real-time API integration with MPLADS live production database
- Mobile native applications (iOS/Android)
- Multi-language NLP (MVP: English only)
- Automated corrective action workflows
- Integration with PFMS (Public Financial Management System) live feeds
- Biometric or identity verification modules
- SMS/WhatsApp notification delivery
- Predictive budget forecasting models (future phase)

1.3 Intended Audience

Audience

Usage

AI Agentic Development System Primary consumer — interprets and executes all requirements

Hackathon Judges (MoSPI/DIID) Evaluation of completeness, feasibility, and innovation

Audience

Solo Developer (Author)

Usage

Reference guide for manual overrides and debugging

1.4 Glossary of Terms

YAML

GLOSSARY:

MPLADS:

full_form: "Members of Parliament Local Area Development Scheme"

definition: "Central Sector Scheme where MPs recommend developmental works for durable community assets and basic civic amenities"

context: "Primary domain of this system"

MP:

full_form: "Member of Parliament"

definition: "Elected representative of Lok Sabha or nominated/elected member of Rajya Sabha"

context: "Primary recommending authority in MPLADS"

MoSPI:

full_form: "Ministry of Statistics and Programme Implementation"

definition: "Central ministry administering MPLADS"

context: "Organization that owns this system"

DIID:

full_form: "Data Informatics & Innovation Division"

definition: "Division within MoSPI responsible for data analytics and technology innovation"

context: "Department commissioning this project"

CONSTITUENCY:

definition: "A parliamentary constituency represented by a Lok Sabha MP, or a state assigned to a Rajya Sabha MP"

context: "Primary geographic unit for fund allocation"

SANCTION:

definition: "Formal approval of a work/project under MPLADS with associated cost estimate"

context: "Marks the beginning of a work's lifecycle"

DISTRICT_AUTHORITY:

definition: "District Collector/Deputy Commissioner responsible for implementing MPLADS works"

context: "Primary implementing authority"

STATE_NODAL_AUTHORITY:

definition: "State-level officer coordinating MPLADS implementation across districts"

context: "Intermediate monitoring authority"

FUND_UTILIZATION_RATE:

definition: "Ratio of expenditure to funds released, expressed as percentage"

formula: $(\text{total_expenditure} / \text{total_funds_released}) * 100$

context: "Key efficiency metric"

COST_OVERRUN:

definition: "When actual expenditure on a work exceeds the sanctioned cost estimate"

formula: $((\text{actual_cost} - \text{sanctioned_cost}) / \text{sanctioned_cost}) * 100$

threshold: ">15% flagged as anomaly in MVP"

DUPLICATE_WORK:

definition: "Two or more sanctioned works with substantially similar descriptions, locations, and cost estimates within the same constituency"

detection_method: "NLP cosine similarity > 0.85 on work descriptions + geographic proximity"

ANOMALY:

definition: "A data point or pattern that deviates significantly from expected norms"

detection_methods: ["z-score", "IQR", "Isolation Forest", "text similarity"]

RISK_SCORE:

definition: "Composite numerical score (0-100) indicating probability and severity of fraud/inefficiency"

tiers:

LOW: "0-25"

MEDIUM: "26-50"

HIGH: "51-75"

CRITICAL: "76-100"

DURABLE_ASSET:

definition: "Physical infrastructure or community facility created under MPLADS that provides lasting benefit"

examples: ["Community halls", "Roads", "School buildings", "Drinking water facilities"]

PFMS:

full_form: "Public Financial Management System"

definition: "Government of India's financial management platform for tracking fund flows"

WCAG:

full_form: "Web Content Accessibility Guidelines"

version: "2.1 AA"

context: "Accessibility compliance standard required for government websites"

NIC:

full_form: "National Informatics Centre"

definition: "Government IT infrastructure and services provider"

context: "Potential deployment infrastructure partner"

GIGW:

full_form: "Guidelines for Indian Government Websites"

definition: "Standards issued by NIC/MeitY for all government web applications"

API:

full_form: "Application Programming Interface"

definition: "Programmatic interface for data exchange between systems"

ETL:

full_form: "Extract, Transform, Load"

definition: "Data pipeline process for ingesting raw data into structured storage"

IQR:

full_form: "Interquartile Range"

definition: "Statistical measure of spread; used here for outlier detection"

formula: " $Q3 - Q1$; outlier if value $< Q1 - 1.5 \cdot IQR$ or $> Q3 + 1.5 \cdot IQR$ "

Z_SCORE:

definition: "Number of standard deviations a data point is from the mean"

formula: " $(x - \mu) / \sigma$ "

threshold: " $|z| > 2.5$ flagged as anomaly"

ISOLATION_FOREST:

definition: "Unsupervised ML algorithm for anomaly detection based on random partitioning"

context: "Used for multivariate anomaly detection on expenditure patterns"

COSINE_SIMILARITY:

definition: "Measure of similarity between two text vectors; 1.0 = identical, 0.0 = orthogonal"

threshold: ">0.85 flags potential duplicate"

JWT:

full_form: "JSON Web Token"

definition: "Token-based authentication standard"

RBAC:

full_form: "Role-Based Access Control"

definition: "Authorization model where permissions are assigned to roles, not individuals"

1.5 References

YAML

REFERENCES:

- id: REF-001

title: "MPLADS Guidelines 2016 (Revised)"

source: "MoSPI"

url: "https://www.mplads.gov.in"

relevance: "Scheme rules, fund allocation norms, work categories"

- id: REF-002

title: "MPLADS Public Dashboard"

source: "MoSPI"

url: "https://mplads.mospi.gov.in/digigov/dashboard.html"

relevance: "Primary data source for MVP"

- id: REF-003

title: "GIGW 3.0 Guidelines"

source: "NIC/MeitY"

relevance: "Web application compliance standards"

- id: REF-004

title: "IT Act 2000 and IT Amendment Act 2008"

source: "Government of India"

relevance: "Legal framework for data handling"

- id: REF-005

title: "Digital Personal Data Protection Act 2023"

source: "Government of India"

relevance: "Data privacy compliance"

SECTION 2: PRODUCT PERSPECTIVE

2.1 System Context within MoSPI Ecosystem

YAML

ECOSYSTEM_MAPPING:

CURRENT_STATE:

existing_system: "MPLADS Online Portal (mplads.mospi.gov.in)"

current_capabilities:

- Manual data entry by District Authorities
- Basic dashboard showing fund release and utilization statistics
- Static reports generation
- Public-facing summary dashboard

current_limitations:

- No automated anomaly detection
- No ML-based fraud identification
- No predictive analytics
- No cross-constituency pattern analysis
- Manual compliance monitoring only
- Limited trend analysis capabilities

PROPOSED_SYSTEM_ROLE:

name: "MPLADS Sentinel"

relationship_to_existing: "Supplementary analytics layer — reads data from existing system, does NOT modify source data"

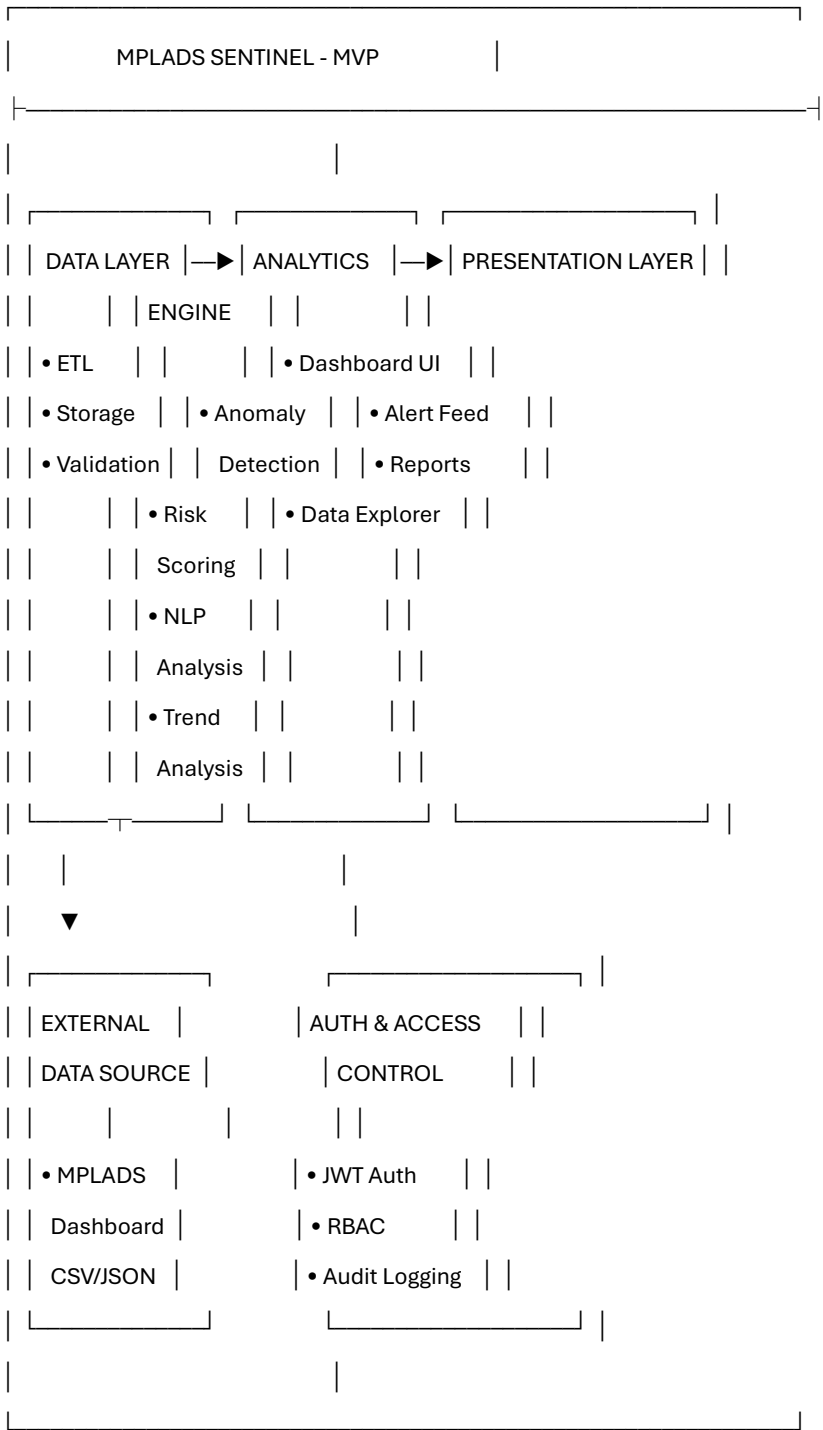
integration_pattern: "Read-only data consumer"

value_addition:

- Automated anomaly detection on existing data
- AI-powered duplicate work identification
- Risk scoring and prioritization
- Predictive trend analysis
- Visual decision-support dashboards
- Automated alert generation

2.2 System Architecture Overview

text



2.3 Interface Specifications

YAML

INTERFACES:

EXTERNAL_DATA_INTERFACE:

id: IF-001

name: "MPLADS Data Ingestion"

type: "File-based + Web Scraping"

source: "https://mplads.mospi.gov.in/digigov/dashboard.html"

data_formats_accepted: ["CSV", "JSON", "XLSX"]

ingestion_methods:

- primary: "Manual CSV upload via admin interface"
- secondary: "Automated web scraping of public dashboard data"
- tertiary: "Synthetic data generator for demo/testing"

refresh_frequency_mvp: "On-demand (manual trigger)"

data_entities:

- constituency_master
- mp_details
- sanctioned_works
- fund_releases
- expenditure_records
- work_progress_status
- payment_details
- asset_creation_records

USER_INTERFACE:

id: IF-002

name: "Web Dashboard"

type: "Browser-based SPA"

protocol: "HTTPS"

supported_browsers:

- "Chrome >= 90"
- "Firefox >= 88"
- "Edge >= 90"
- "Safari >= 14"

responsive_breakpoints:

- mobile: "320px - 767px"
- tablet: "768px - 1023px"
- desktop: "1024px+"

accessibility: "WCAG 2.1 AA compliant"

API_INTERFACE:

id: IF-003

name: "REST API"

type: "RESTful JSON API"

protocol: "HTTPS"

authentication: "JWT Bearer Token"

rate_limiting: "100 requests/minute per authenticated user"

versioning: "URL path versioning (/api/v1/)"

documentation: "OpenAPI 3.0 Swagger specification auto-generated"

DATABASE_INTERFACE:

id: IF-004

name: "Data Storage"

type: "PostgreSQL 15+ with pgvector extension"

connection: "Internal only, no external access"

encryption: "AES-256 at rest, TLS 1.3 in transit"

2.4 Smart Automation Theme Alignment

YAML

SMART_AUTOMATION_ALIGNMENT:

THEME: "Smart Automation"

ALIGNMENT_POINTS:

- area: "Automated Monitoring"
 - description: "Replaces manual review of thousands of MPLADS works with AI-driven continuous monitoring"
 - automation_gain: "Estimated 90% reduction in manual monitoring effort for anomaly identification"

- area: "Intelligent Alerting"
 - description: "Proactive alert generation instead of reactive complaint-driven investigation"
 - automation_gain: "Near-real-time anomaly flagging vs current quarterly manual review cycles"

- area: "Data-Driven Decision Making"
 - description: "ML-powered risk scores and trend analysis replace subjective assessment"
 - automation_gain: "Quantified risk scoring enables objective prioritization of investigations"

- area: "Pattern Recognition at Scale"
 - description: "NLP-based duplicate detection across 543+ constituencies simultaneously"
 - automation_gain: "Cross-constituency pattern detection impossible with manual methods"

- area: "Predictive Intelligence"
 - description: "Trend analysis enables early warning before issues become critical"
 - automation_gain: "Shift from post-hoc auditing to preventive monitoring"

2.5 User Roles and Personas

YAML

USER_ROLES:

- role_id: ROLE_ADMIN

name: "System Administrator"

description: "MoSPI/DIID technical staff managing the platform"

permissions:

- data_upload

- user_management

- system_configuration

- all_data_access

- alert_management

- report_generation

dashboard_view: "Full system overview + administration panel"

- role_id: ROLE_MINISTRY

name: "Ministry Official"

description: "MoSPI officials monitoring national MPLADS implementation"

permissions:

- national_dashboard_view

- all_constituency_data_read

- alert_view

- report_generation

- trend_analysis

dashboard_view: "National overview with drill-down capability"

- role_id: ROLE_STATE_NODAL

name: "State Nodal Authority"

description: "State-level officers coordinating MPLADS within their state"

permissions:

- state_dashboard_view

- state_constituency_data_read

- alert_view_state

- report_generation_state

dashboard_view: "State-filtered view with constituency drill-down"

- role_id: ROLE_DISTRICT

name: "District Authority"

description: "District Collector/DC implementing MPLADS works"

permissions:

- district_dashboard_view
- district_data_read
- alert_view_district
- report_generation_district

dashboard_view: "District-specific work and expenditure view"

- role_id: ROLE_MP

name: "Member of Parliament"

description: "MP viewing their constituency's MPLADS data and insights"

permissions:

- constituency_dashboard_view
- constituency_data_read
- alert_view_constituency

dashboard_view: "Constituency-specific simplified view"

- role_id: ROLE_PUBLIC

name: "Public Viewer"

description: "Citizens accessing transparency dashboard"

permissions:

- public_dashboard_view
- aggregate_statistics_read

dashboard_view: "Anonymized aggregate statistics only"

SECTION 3: FUNCTIONAL REQUIREMENTS

3.1 Feature Module: Data Ingestion & Management (DIM)

FR-DIM-001: CSV/XLSX Data Upload

YAML

REQUIREMENT:

id: FR-DIM-001

title: "Structured Data File Upload"

priority: MUST_HAVE

module: DATA_INGESTION

USER STORY: >

AS a System Administrator,
I WANT to upload CSV/XLSX files containing MPLADS data
(sanctions, expenditures, works, payments, fund releases),
SO THAT the system can ingest and store structured MPLADS data
for analysis.

INPUT_SPECIFICATION:

file_formats: ["CSV (UTF-8)", "XLSX"]

max_file_size: "50 MB per file"

required_data_schemas:

- schema_name: "sanctioned_works"

required_columns:

- column: "work_id"

type: "string"

constraints: "unique, not null"

- column: "constituency_name"

type: "string"

constraints: "not null"

- column: "state_name"

type: "string"

constraints: "not null"

- column: "district_name"

type: "string"

constraints: "not null"

- column: "mp_name"

type: "string"

constraints: "not null"

- column: "work_description"

type: "string"

constraints: "not null, min_length=10"

- column: "work_category"

type: "string"

constraints: "not null, enum: [EDUCATION, HEALTH, DRINKING_WATER, SANITATION, ROADS, COMMUNITY_ASSETS, POWER, SPORTS, OTHER]"

- column: "sanctioned_amount"

type: "decimal"

constraints: "not null, > 0, max 2 decimal places, unit: INR"

- column: "sanction_date"
type: "date"
constraints: "not null, format: YYYY-MM-DD"
- column: "expected_completion_date"
type: "date"
constraints: "format: YYYY-MM-DD"
- column: "actual_expenditure"
type: "decimal"
constraints: ">= 0, unit: INR"
- column: "work_status"
type: "string"
constraints: "enum: [SANCTIONED, IN_PROGRESS, COMPLETED, CANCELLED, ON_HOLD]"
- column: "completion_date"
type: "date"
constraints: "format: YYYY-MM-DD, nullable"
- column: "implementing_agency"
type: "string"
constraints: "not null"
- column: "financial_year"
type: "string"
constraints: "not null, format: YYYY-YY (e.g., 2023-24)"
- column: "latitude"
type: "decimal"
constraints: "nullable, range: 6.0 to 37.0"
- column: "longitude"
type: "decimal"
constraints: "nullable, range: 68.0 to 97.5"

- schema_name: "fund_releases"

required_columns:

- column: "release_id"
type: "string"
constraints: "unique, not null"
- column: "constituency_name"
type: "string"
constraints: "not null"
- column: "financial_year"

type: "string"
constraints: "not null"
- column: "installment_number"
type: "integer"
constraints: "not null, enum: [1, 2]"
- column: "amount_released"
type: "decimal"
constraints: "not null, > 0, unit: INR"
- column: "release_date"
type: "date"
constraints: "not null, format: YYYY-MM-DD"
- column: "cumulative_release"
type: "decimal"
constraints: ">= 0, unit: INR"

OUTPUT_SPECIFICATION:

success_response:

status: 200

body:

upload_id: "string (UUID)"
records_parsed: "integer"
records_valid: "integer"
records_rejected: "integer"
validation_errors: "array of {row_number, column, error_message}"
ingestion_timestamp: "ISO 8601 datetime"

error_response:

status: 400

body:

error_code: "string"
error_message: "string"
details: "array of validation failures"

PROCESSING_RULES:

- "Validate file format before parsing"
- "Validate each row against schema constraints"
- "Reject individual invalid rows; ingest valid rows"
- "Generate upload audit log entry with user_id, timestamp, file_hash, result_summary"

- "Deduplicate against existing records using work_id as primary key"
- "On duplicate work_id: update if newer data, log conflict"
- "Trigger anomaly detection pipeline on successful ingestion of ≥ 1 valid record"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DIM-001-01

description: "System accepts CSV file with valid schema and ingests all valid rows"

test: "Upload CSV with 100 valid rows $\rightarrow$ records_valid == 100, records_rejected == 0"

- criteria_id: AC-DIM-001-02

description: "System rejects rows with missing required fields"

test: "Upload CSV where row 5 has null work_id $\rightarrow$ records_rejected ≥ 1 , validation_errors includes {row: 5, column: work_id}"

- criteria_id: AC-DIM-001-03

description: "System rejects file exceeding 50MB"

test: "Upload 55MB CSV $\rightarrow$ HTTP 413, error_message contains 'file size exceeds limit'"

- criteria_id: AC-DIM-001-04

description: "System handles duplicate work_ids by updating existing records"

test: "Upload CSV with work_id='W001' already in DB with different expenditure $\rightarrow$ DB record updated, audit log entry created"

- criteria_id: AC-DIM-001-05

description: "Upload creates audit trail entry"

test: "After upload $\rightarrow$ audit_log table contains entry with upload_id, user_id, timestamp, file_hash"

FR-DIM-002: Synthetic Data Generator

YAML

REQUIREMENT:

id: FR-DIM-002

title: "Synthetic MPLADS Data Generator for Demo/Testing"

priority: MUST_HAVE

module: DATA_INGESTION

USER STORY: >

AS an AI development system,

I WANT to generate realistic synthetic MPLADS data including known anomalies,

SO THAT the anomaly detection system can be demonstrated and validated

without requiring actual government data.

INPUT_SPECIFICATION:

parameters:

num_constituencies:

type: "integer"

default: 50

range: "10 to 543"

num_works_per_constituency:

type: "integer"

default: 100

range: "20 to 500"

financial_years:

type: "array of string"

default: ["2019-20", "2020-21", "2021-22", "2022-23", "2023-24"]

anomaly_injection_rate:

type: "decimal"

default: 0.08

description: "Fraction of records with injected anomalies"

anomaly_types:

type: "array"

values:

- type: "COST_OVERRUN"

injection_method: "actual_expenditure = sanctioned_amount * random(1.2, 3.0)"

label: true

- type: "DUPLICATE_WORK"

injection_method: "Copy work_description with minor word substitution, same constituency, similar amount"

label: true

- type: "DELAYED_PROJECT"

injection_method: "completion_date = expected_completion_date + random(180, 730) days; OR status IN_PROGRESS with expected_completion_date > 365 days ago"

label: true

- type: "ABNORMAL_EXPENDITURE_PATTERN"

injection_method: "Single constituency with fund_utilization_rate < 20% or > 150%"

label: true

- type: "SUSPICIOUS_CLUSTERING"

injection_method: "Multiple works with identical amounts in same constituency within same month"

label: true

OUTPUT_SPECIFICATION:

files_generated:

- "synthetic_sanctioned_works.csv"
- "synthetic_fund_releases.csv"
- "synthetic_anomaly_labels.csv"

anomaly_labels_schema:

columns:

- "work_id"
- "anomaly_type"
- "anomaly_injected (boolean)"
- "anomaly_description"

PROCESSING_RULES:

- "Use realistic Indian state/district/constituency names from official list"
- "Generate work descriptions using template-based NLG with category-appropriate vocabulary"
- "Ensure sanctioned amounts follow realistic distribution: median ~₹5 lakhs, range ₹50,000 to ₹50 lakhs"
- "Normal fund utilization rate: 60-95%"
- "Ensure date consistency: sanction_date < expected_completion_date; completion_date >= sanction_date"
- "Generate ground-truth anomaly labels for validation"
- "Set random seed for reproducibility: seed=42"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DIM-002-01

description: "Generator produces valid CSV files matching sanctioned_works schema"

test: "Run generator with defaults → output CSV passes schema validation from FR-DIM-001"

- criteria_id: AC-DIM-002-02

description: "Anomaly injection rate is approximately as specified"

test: "With anomaly_injection_rate=0.08 → |actual_anomaly_count/total_records - 0.08| < 0.02"

- criteria_id: AC-DIM-002-03

description: "Ground truth labels are accurate"

test: "Every record in synthetic_anomaly_labels.csv with anomaly_injected=true has corresponding anomalous values in the data"

- criteria_id: AC-DIM-002-04

description: "Data is reproducible with same seed"

test: "Two runs with seed=42 produce byte-identical CSV output"

FR-DIM-003: Data Scraping from MPLADS Dashboard

YAML

REQUIREMENT:

id: FR-DIM-003

title: "Web Scraping of Public MPLADS Dashboard Data"

priority: SHOULD_HAVE

module: DATA_INGESTION

USER_STORY: >

AS a System Administrator,

I WANT the system to scrape publicly available MPLADS data from the official dashboard,

SO THAT the analytics engine has access to the latest published data.

INPUT_SPECIFICATION:

target_url: "https://mplads.mospi.gov.in/digigov/dashboard.html"

scraping_targets:

- "State-wise fund release and utilization summary"
- "Constituency-wise work statistics"
- "Year-wise sanction and expenditure data"
- "Work category distribution"

trigger: "Manual trigger by ROLE_ADMIN via admin panel button"

OUTPUT_SPECIFICATION:

output_format: "Structured JSON/CSV stored in database"

success_response:

records_scraped: "integer"

scrape_timestamp: "ISO 8601 datetime"

data_freshness: "Date of latest data point scraped"

PROCESSING_RULES:

- "Respect robots.txt directives"
- "Add 2-second delay between requests to avoid overloading server"
- "Parse HTML tables / JavaScript-rendered content using headless browser if needed"
- "Map scraped fields to internal schema"

- "Log all scraping activity with timestamp and result"
- "Handle scraping failures gracefully with retry (max 3 attempts)"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DIM-003-01
description: "Scraper retrieves at least state-wise summary data"
test: "Trigger scrape → records_scraped > 0 for state-wise fund data"
- criteria_id: AC-DIM-003-02
description: "Scraped data conforms to internal schema"
test: "All scraped records pass schema validation"
- criteria_id: AC-DIM-003-03
description: "Scraper handles site unavailability gracefully"
test: "Simulate HTTP 500 → system returns error message, no crash, logged appropriately"

3.2 Feature Module: Anomaly Detection Engine (ADE)

FR-ADE-001: Cost Overrun Detection

YAML

REQUIREMENT:

id: FR-ADE-001
title: "Cost Overrun Anomaly Detection"
priority: MUST_HAVE
module: ANOMALY_DETECTION

USER_STORY: >

AS a Ministry Official,
I WANT the system to automatically detect works where actual expenditure significantly exceeds the sanctioned cost estimate,
SO THAT I can investigate potential financial irregularities.

INPUT_SPECIFICATION:

data_source: "sanctioned_works table"
required_fields:

- "work_id"
- "sanctioned_amount"
- "actual_expenditure"
- "work_category"

- "constituency_name"
- "financial_year"

DETECTION_ALGORITHM:

method_1_rule_based:

name: "Threshold-based cost overrun"

logic: |

$\text{cost_overrun_percentage} = ((\text{actual_expenditure} - \text{sanctioned_amount}) / \text{sanctioned_amount}) * 100$

IF $\text{cost_overrun_percentage} > 15\%$ THEN flag as COST_OVERRUN anomaly

severity:

MEDIUM: $15\% < \text{cost_overrun_percentage} \leq 30\%$

HIGH: $30\% < \text{cost_overrun_percentage} \leq 50\%$

CRITICAL: $\text{cost_overrun_percentage} > 50\%$

method_2_statistical:

name: "Z-score based outlier detection"

logic: |

Group works by work_category

For each category:

Calculate mean and std_dev of cost_overrun_percentage

$\text{z_score} = (\text{cost_overrun_percentage} - \text{mean}) / \text{std_dev}$

IF $|\text{z_score}| > 2.5$ THEN flag as statistical anomaly

Combine with method_1 for composite detection

method_3_ml:

name: "Isolation Forest on expenditure features"

logic: |

Features: [sanctioned_amount, actual_expenditure, cost_overrun_percentage,
days_to_completion, work_category_encoded]

Train Isolation Forest with contamination=0.05

Flag records with anomaly_score < threshold as anomalies

training_trigger: "On each data ingestion with ≥ 500 records"

OUTPUT_SPECIFICATION:

anomaly_record:

work_id: "string"

anomaly_type: "COST_OVERRUN"

severity: "MEDIUM | HIGH | CRITICAL"
cost_overrun_percentage: "decimal"
sanctioned_amount: "decimal"
actual_expenditure: "decimal"
detection_method: "string"
confidence_score: "decimal (0.0 to 1.0)"
detected_at: "ISO 8601 datetime"
constituency: "string"
state: "string"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-ADE-001-01
description: "System flags works with >15% cost overrun"
test: "Insert work with sanctioned=100000, actual=120000 → flagged with severity MEDIUM"

- criteria_id: AC-ADE-001-02
description: "System correctly assigns severity tiers"
test: |
sanctioned=100000, actual=140000 → HIGH
sanctioned=100000, actual=160000 → CRITICAL

- criteria_id: AC-ADE-001-03
description: "Statistical method detects contextual anomalies"
test: "Insert EDUCATION work with 10% overrun where category mean is 2% → flagged by z-score method"

- criteria_id: AC-ADE-001-04
description: "System does not flag normal works as cost overrun"
test: "Insert work with sanctioned=100000, actual=90000 → NOT flagged"

- criteria_id: AC-ADE-001-05
description: "On synthetic data with ground truth, precision ≥ 0.7 and recall ≥ 0.8 for cost overrun class"
test: "Run detection on synthetic data → compute precision/recall against labels"

FR-ADE-002: Duplicate Work Detection

YAML

REQUIREMENT:

id: FR-ADE-002
title: "Duplicate Work Detection using NLP"

priority: MUST_HAVE

module: ANOMALY_DETECTION

USER_STORY: >

AS a District Authority,

I WANT the system to identify potentially duplicate works

(similar descriptions, same area, similar costs),

SO THAT I can prevent double-funding of the same project.

INPUT_SPECIFICATION:

data_source: "sanctioned_works table"

required_fields:

- "work_id"
- "work_description"
- "constituency_name"
- "district_name"
- "sanctioned_amount"
- "sanction_date"
- "work_category"
- "latitude (optional)"
- "longitude (optional)"

DETECTION_ALGORITHM:

step_1_text_embedding:

method: "Sentence transformer model (all-MiniLM-L6-v2 or equivalent)"

process: |

For each work_description:

Generate 384-dimensional embedding vector

Store in pgvector column for efficient similarity search

step_2_similarity_search:

method: "Cosine similarity with ANN (Approximate Nearest Neighbor)"

process: |

For each work in a constituency:

Find all works in SAME constituency with cosine_similarity > 0.85

Filter pairs where:

- $|\text{sanctioned_amount_A} - \text{sanctioned_amount_B}| / \max(A, B) < 0.3$ (amount within 30%)

- $|\text{sanction_date_A} - \text{sanction_date_B}| < 365 \text{ days}$
 - $\text{work_category_A} == \text{work_category_B}$ (optional, boosts confidence)
- IF geographic coords available:
- $\text{haversine_distance}(\text{A}, \text{B}) < 2 \text{ km}$ (boosts confidence)

step_3_scoring:

composite_duplicate_score: |

score = 0

score += text_similarity * 40 (max 40 points)

score += amount_similarity * 20 (max 20 points)

score += temporal_proximity * 15 (max 15 points, linear decay over 365 days)

score += geographic_proximity * 15 (max 15 points, if coords available)

score += category_match * 10 (10 if same category, 0 otherwise)

IF score ≥ 70 : flag as POTENTIAL_DUPLICATE, severity HIGH

IF score ≥ 50 AND score < 70 : flag as POSSIBLE_DUPLICATE, severity MEDIUM

IF score < 50 : do not flag

OUTPUT_SPECIFICATION:

duplicate_pair_record:

pair_id: "string (UUID)"

work_id_a: "string"

work_id_b: "string"

text_similarity: "decimal (0.0 to 1.0)"

amount_similarity: "decimal (0.0 to 1.0)"

composite_score: "integer (0 to 100)"

severity: "MEDIUM | HIGH"

anomaly_type: "DUPLICATE_WORK"

constituency: "string"

detected_at: "ISO 8601 datetime"

work_description_a: "string"

work_description_b: "string"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-ADE-002-01

description: "System detects near-identical work descriptions"

test: |

Insert: "Construction of community hall at Village Rampur" and

"Construction of community hall in Village Rampur"

Same constituency, similar amount → flagged as POTENTIAL_DUPLICATE

- criteria_id: AC-ADE-002-02

description: "System does NOT flag genuinely different works"

test: |

Insert: "Construction of primary school building" and

"Installation of solar street lights"

Same constituency → NOT flagged

- criteria_id: AC-ADE-002-03

description: "System does NOT flag similar works in different constituencies"

test: |

Insert: identical descriptions in different constituencies → NOT flagged

(cross-constituency duplicates are out of MVP scope)

- criteria_id: AC-ADE-002-04

description: "Detection runs within 30 seconds for 5000 works in one constituency"

test: "Performance benchmark on synthetic data"

- criteria_id: AC-ADE-002-05

description: "On synthetic data, precision ≥ 0.75 and recall ≥ 0.80 for duplicate class"

test: "Validate against ground truth labels"

FR-ADE-003: Delayed Project Detection

YAML

REQUIREMENT:

id: FR-ADE-003

title: "Delayed Project Identification"

priority: MUST_HAVE

module: ANOMALY_DETECTION

USER_STORY: >

AS a State Nodal Authority,

I WANT the system to identify projects that are significantly delayed

beyond their expected completion date,

SO THAT I can escalate and take corrective action.

INPUT_SPECIFICATION:

data_source: "sanctioned_works table"

required_fields:

- "work_id"
- "sanction_date"
- "expected_completion_date"
- "completion_date (nullable)"
- "work_status"
- "constituency_name"

reference_date: "Current system date (or configurable analysis date)"

DETECTION_ALGORITHM:

logic: |

FOR each work:

IF work_status IN ['SANCTIONED', 'IN_PROGRESS', 'ON_HOLD']:

IF expected_completion_date IS NOT NULL AND reference_date > expected_completion_date:

delay_days = (reference_date - expected_completion_date).days

IF delay_days > 90 AND delay_days <= 180: severity = MEDIUM

IF delay_days > 180 AND delay_days <= 365: severity = HIGH

IF delay_days > 365: severity = CRITICAL

Flag as DELAYED_PROJECT

IF work_status == 'COMPLETED' AND completion_date IS NOT NULL:

IF completion_date > expected_completion_date:

delay_days = (completion_date - expected_completion_date).days

IF delay_days > 180: flag as DELAYED_PROJECT (completed but was severely delayed)

IF work_status == 'SANCTIONED' AND (reference_date - sanction_date).days > 365:

Flag as STALLED_PROJECT, severity HIGH

(Sanctioned over a year ago, never started)

OUTPUT_SPECIFICATION:

anomaly_record:

work_id: "string"

anomaly_type: "DELAYED_PROJECT | STALLED_PROJECT"

severity: "MEDIUM | HIGH | CRITICAL"

delay_days: "integer"
sanction_date: "date"
expected_completion_date: "date"
current_status: "string"
constituency: "string"
detected_at: "ISO 8601 datetime"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-ADE-003-01

description: "Work delayed 100 days beyond expected completion is flagged MEDIUM"

test: "expected_completion=2024-01-01, reference_date=2024-04-11, status=IN_PROGRESS → DELAYED_PROJECT, MEDIUM"

- criteria_id: AC-ADE-003-02

description: "Work delayed 400 days is flagged CRITICAL"

test: "expected_completion=2023-01-01, reference_date=2024-03-06, status=IN_PROGRESS → CRITICAL"

- criteria_id: AC-ADE-003-03

description: "Completed work is not flagged if completed before deadline"

test: "expected_completion=2024-06-01, completion_date=2024-05-01, status=COMPLETED → NOT flagged"

- criteria_id: AC-ADE-003-04

description: "Sanctioned work with no progress after 1 year flagged as STALLED"

test: "sanction_date=2023-01-01, status=SANCTIONED, reference_date=2024-06-01 → STALLED_PROJECT, HIGH"

FR-ADE-004: Fund Utilization Anomaly Detection

YAML

REQUIREMENT:

id: FR-ADE-004
title: "Abnormal Fund Utilization Pattern Detection"
priority: MUST_HAVE
module: ANOMALY_DETECTION

USER STORY: >

AS a Ministry Official,

I WANT to identify constituencies with abnormally low or high fund utilization rates,

SO THAT I can investigate potential mismanagement or fund diversion.

INPUT_SPECIFICATION:

data_sources:

- "fund_releases table"
- "sanctioned_works table (aggregated expenditure)"

required_computed_metrics:

- fund_utilization_rate: $(\text{total_expenditure} / \text{total_funds_released}) * 100$
- works_per_crore: $\text{total_works_sanctioned} / (\text{total_funds_released} / 10000000)$
- average_work_cost: $\text{total_sanctioned_amount} / \text{total_works_sanctioned}$

grouping: "Per constituency per financial_year"

DETECTION_ALGORITHM:

method_1_threshold:

logic: |

IF fund_utilization_rate < 30%: flag LOW_UTILIZATION, severity HIGH

IF fund_utilization_rate < 50%: flag LOW_UTILIZATION, severity MEDIUM

IF fund_utilization_rate > 110%: flag OVER_UTILIZATION, severity HIGH (spending more than released)

method_2_statistical:

logic: |

Group all constituencies by state

For each state:

Calculate mean and std of fund_utilization_rate

$z_score = (\text{constituency_rate} - \text{state_mean}) / \text{state_std}$

IF $|z_score| > 2.0$: flag as FUND_UTILIZATION_ANOMALY

method_3_temporal:

logic: |

For each constituency:

Calculate year-over-year change in fund_utilization_rate

IF $|change| > 40$ percentage points: flag as SUDDEN_UTILIZATION_SHIFT

OUTPUT_SPECIFICATION:

anomaly_record:

constituency: "string"

financial_year: "string"

anomaly_type: "LOW_UTILIZATION | OVER_UTILIZATION | SUDDEN_UTILIZATION_SHIFT"

severity: "MEDIUM | HIGH | CRITICAL"

fund_utilization_rate: "decimal"
total_funds_released: "decimal"
total_expenditure: "decimal"
state_average_rate: "decimal"
z_score: "decimal"
detected_at: "ISO 8601 datetime"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-ADE-004-01
description: "Constituency with 25% utilization rate flagged as LOW_UTILIZATION HIGH"
test: "released=5000000, expenditure=1250000 → LOW_UTILIZATION, HIGH"
- criteria_id: AC-ADE-004-02
description: "Constituency with 85% utilization NOT flagged"
test: "released=5000000, expenditure=4250000 → NOT flagged by threshold method"
- criteria_id: AC-ADE-004-03
description: "Sudden drop from 90% to 30% year-over-year flagged"
test: "FY 2022-23: 90%, FY 2023-24: 30% → SUDDEN_UTILIZATION_SHIFT flagged"

FR-ADE-005: Suspicious Pattern Detection

YAML

REQUIREMENT:

id: FR-ADE-005
title: "Suspicious Expenditure Pattern Detection"
priority: SHOULD_HAVE
module: ANOMALY_DETECTION

USER_STORY: >

AS a Ministry Official,
I WANT the system to detect suspicious patterns such as
clustering of identical amounts, end-of-year spending spikes,
and unusual implementing agency patterns,
SO THAT potential manipulation or gaming of the system can be identified.

DETECTION_PATTERNS:

pattern_1_amount_clustering:
description: "Multiple works with identical sanctioned amounts in same constituency"

logic: |

Group works by (constituency, financial_year)

Count occurrences of each unique sanctioned_amount

IF count ≥ 5 for same amount: flag as AMOUNT_CLUSTERING

severity: MEDIUM if 5-10 occurrences, HIGH if > 10

pattern_2_end_of_year_rush:

description: "Disproportionate sanctions in last month of financial year"

logic: |

For each constituency per financial_year:

Count works sanctioned in March (last month)

Calculate ratio = march_sanctions / total_annual_sanctions

IF ratio > 0.40 : flag as END_OF_YEAR_RUSH, severity MEDIUM

IF ratio > 0.60 : severity HIGH

pattern_3_round_number_bias:

description: "Excessive use of round numbers in sanctioned amounts"

logic: |

For each constituency:

Calculate percentage of works with sanctioned_amount divisible by 100000

IF percentage $> 80\%$: flag as ROUND_NUMBER_BIAS, severity LOW

(Informational flag — not directly fraud but indicates potential estimate padding)

pattern_4_single_agency_dominance:

description: "One implementing agency receiving disproportionate share of works"

logic: |

For each constituency per financial_year:

Calculate each agency's share of total works and total amount

IF any agency $> 80\%$ of total amount: flag as AGENCY_CONCENTRATION, severity MEDIUM

OUTPUT_SPECIFICATION:

anomaly_record:

constituency: "string"

financial_year: "string"

anomaly_type: "AMOUNT_CLUSTERING | END_OF_YEAR_RUSH | ROUND_NUMBER_BIAS |
AGENCY_CONCENTRATION"

severity: "LOW | MEDIUM | HIGH"

details: "JSON object with pattern-specific details"

detected_at: "ISO 8601 datetime"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-ADE-005-01

description: "7 works with identical amount ₹5,00,000 in same constituency flagged"

test: "Insert 7 works with sanctioned_amount=500000 → AMOUNT_CLUSTERING, MEDIUM"

- criteria_id: AC-ADE-005-02

description: "55% of sanctions in March flagged as END_OF_YEAR_RUSH"

test: "Constituency with 20 works, 11 in March → flagged MEDIUM"

3.3 Feature Module: Risk Scoring Engine (RSE)

FR-RSE-001: Composite Risk Score Calculation

YAML

REQUIREMENT:

id: FR-RSE-001

title: "Composite Risk Score per Work Item and per Constituency"

priority: MUST_HAVE

module: RISK_SCORING

USER_STORY: >

AS a Ministry Official,

I WANT each MPLADS work and each constituency to have a composite risk score,

SO THAT I can prioritize investigation efforts on the highest-risk items.

SCORING_MODEL:

work_level_score:

max_score: 100

components:

- component: "cost_overrun_score"

weight: 25

calculation: |

IF no cost overrun: 0

IF overrun 0-15%: 5

IF overrun 15-30%: 15

IF overrun 30-50%: 20

IF overrun > 50%: 25

- component: "delay_score"

weight: 25

calculation: |

IF not delayed: 0

IF delayed 0-90 days: 5

IF delayed 90-180 days: 10

IF delayed 180-365 days: 20

IF delayed > 365 days: 25

- component: "duplicate_score"

weight: 25

calculation: |

IF no duplicate detected: 0

IF possible duplicate (score 50-70): 15

IF probable duplicate (score > 70): 25

- component: "pattern_anomaly_score"

weight: 15

calculation: |

Sum of active pattern flags:

AMOUNT_CLUSTERING: +5

END_OF_YEAR_RUSH: +5

ROUND_NUMBER_BIAS: +2

AGENCY_CONCENTRATION: +3

Cap at 15

- component: "fund_utilization_anomaly_score"

weight: 10

calculation: |

IF constituency fund_utilization_rate anomaly detected: 10

ELSE: 0

constituency_level_score:

max_score: 100

calculation: |

avg_work_risk = mean(work_level_scores for all works in constituency)

$\text{high_risk_ratio} = \text{count}(\text{works with score} > 50) / \text{total_works}$

$\text{fund_util_anomaly} = 1 \text{ if any fund utilization anomaly else } 0$

$\text{constituency_score} = (\text{avg_work_risk} * 0.4) + (\text{high_risk_ratio} * 100 * 0.35) + (\text{fund_util_anomaly} * 25)$

Cap at 100

risk_tier_mapping:

LOW: "0-25"

MEDIUM: "26-50"

HIGH: "51-75"

CRITICAL: "76-100"

OUTPUT_SPECIFICATION:

work_risk_record:

work_id: "string"

risk_score: "integer (0-100)"

risk_tier: "LOW | MEDIUM | HIGH | CRITICAL"

component_scores:

cost_overrun: "integer"

delay: "integer"

duplicate: "integer"

pattern: "integer"

fund_utilization: "integer"

last_calculated: "ISO 8601 datetime"

constituency_risk_record:

constituency: "string"

state: "string"

risk_score: "integer (0-100)"

risk_tier: "LOW | MEDIUM | HIGH | CRITICAL"

total_works: "integer"

high_risk_works_count: "integer"

last_calculated: "ISO 8601 datetime"

PROCESSING_RULES:

- "Recalculate scores whenever new anomalies are detected"
- "Store historical scores for trend analysis"

- "Scores must be deterministic given same input data"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-RSE-001-01

description: "Work with 40% cost overrun + 200 days delay scores > 50"

test: "cost_overrun=40%, delay=200 days, no duplicate, no patterns → score = 20 + 20 + 0 + 0 + 0 = 40. Wait — recalculating: overrun 30-50% → 20, delay 180-365 → 20, total = 40 → MEDIUM. Test passes."

- criteria_id: AC-RSE-001-02

description: "Work with no anomalies scores 0"

test: "No overrun, no delay, no duplicate, no pattern → score = 0, tier = LOW"

- criteria_id: AC-RSE-001-03

description: "Work flagged as duplicate with 60% overrun scores CRITICAL"

test: "overrun >50% → 25, duplicate probable → 25 = 50 minimum → HIGH or above"

- criteria_id: AC-RSE-001-04

description: "Constituency score correctly aggregates work scores"

test: "Constituency with 10 works, 5 scoring >50 → high_risk_ratio=0.5, constituency_score > 50"

3.4 Feature Module: Dashboard & Visualization (DVZ)

FR-DVZ-001: National Overview Dashboard

YAML

REQUIREMENT:

id: FR-DVZ-001

title: "National Overview Dashboard with Heatmap"

priority: MUST_HAVE

module: DASHBOARD

USER STORY: >

AS a Ministry Official,

I WANT to see a national overview dashboard showing risk distribution

across all states and constituencies on an interactive map,

SO THAT I can quickly identify geographic clusters of concern.

UI_SPECIFICATION:

page_route: "/dashboard/national"

layout:

header:

- "MPLADS Sentinel" logo and title
- User role badge
- Global date range filter (financial year selector)
- Logout button

section_1_kpi_cards:

position: "Top row, 4 cards equally spaced"

cards:

- id: "kpi_total_works"

label: "Total Works Analyzed"

value_source: "COUNT(sanctioned_works)"

format: "Integer with commas (e.g., 12,345)"

icon: "briefcase"

color: "blue"

- id: "kpi_total_expenditure"

label: "Total Expenditure"

value_source: "SUM(actual_expenditure)"

format: "₹ X.XX Cr (converted to crores, 2 decimal places)"

icon: "currency-rupee"

color: "green"

- id: "kpi_anomalies_detected"

label: "Anomalies Detected"

value_source: "COUNT(anomalies WHERE status=ACTIVE)"

format: "Integer with commas"

icon: "alert-triangle"

color: "red"

- id: "kpi_high_risk_constituencies"

label: "High-Risk Constituencies"

value_source: "COUNT(constituencies WHERE risk_tier IN [HIGH, CRITICAL])"

format: "Integer"

icon: "map-pin"

color: "orange"

section_2_map:

position: "Below KPI cards, full width"

height: "500px"

type: "India choropleth map (state-level heatmap)"

implementation: "Leaflet.js or D3.js with India GeoJSON"

data_binding: "State-level average risk score → color gradient"

color_scale:

low: "#4CAF50 (green, risk 0-25)"

medium: "#FFC107 (amber, risk 26-50)"

high: "#FF9800 (orange, risk 51-75)"

critical: "#F44336 (red, risk 76-100)"

interactions:

- "Hover: Tooltip showing state name, avg risk score, total works, anomaly count"
- "Click: Navigate to state drill-down view (/dashboard/state/{state_id})"

section_3_charts:

position: "Below map, two columns"

chart_left:

type: "Bar chart"

title: "Top 10 High-Risk Constituencies"

data: "Top 10 constituencies by risk_score DESC"

x_axis: "Constituency name"

y_axis: "Risk score (0-100)"

bar_color: "Color-coded by risk tier"

interaction: "Click bar → navigate to constituency detail"

chart_right:

type: "Donut chart"

title: "Anomaly Type Distribution"

data: "COUNT of anomalies grouped by anomaly_type"

segments: ["COST_OVERRUN", "DUPLICATE_WORK", "DELAYED_PROJECT", "FUND_MISUTILIZATION", "PATTERN_ANOMALY"]

colors: "Distinct palette per type"

interaction: "Click segment → filter alert feed below"

section_4_trend:

position: "Below charts, full width"

type: "Line chart (multi-series)"
title: "Anomaly Trends Over Financial Years"
x_axis: "Financial year"
y_axis: "Number of anomalies detected"
series: "One line per anomaly_type"
interaction: "Toggle series visibility"

section_5_alert_feed:

position: "Below trend chart"
type: "Scrollable table/list"
title: "Recent Alerts"
columns:
- "Alert ID"
- "Timestamp"
- "Constituency"
- "State"
- "Anomaly Type (with icon)"
- "Severity (color-coded badge)"
- "Work ID (link to detail)"
- "Brief Description"
default_sort: "Timestamp DESC"
pagination: "20 per page"
filters:
- "Severity: ALL | CRITICAL | HIGH | MEDIUM | LOW"
- "Anomaly Type: multi-select"
- "State: dropdown"
interaction: "Click row → expand alert detail panel"

ACCESSIBILITY_REQUIREMENTS:

- "All charts must have ARIA labels and role='img'"
- "Color-coded elements must also use patterns/icons for color-blind users"
- "Map must be keyboard navigable"
- "KPI cards must be screen-reader friendly with proper heading hierarchy"
- "Minimum contrast ratio 4.5:1 for all text"
- "Focus indicators visible on all interactive elements"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DVZ-001-01

description: "Dashboard loads within 3 seconds with 10,000 works in database"

test: "Measure page load time with 10K records → < 3000ms"

- criteria_id: AC-DVZ-001-02

description: "All 4 KPI cards display correct aggregated values"

test: "Compare displayed values against SQL COUNT/SUM queries → exact match"

- criteria_id: AC-DVZ-001-03

description: "Map renders all 28 states + 8 UTs with correct color coding"

test: "Visual verification + programmatic check that all state GeoJSON features are rendered"

- criteria_id: AC-DVZ-001-04

description: "State click navigates to state drill-down"

test: "Click Maharashtra on map → URL changes to /dashboard/state/maharashtra"

- criteria_id: AC-DVZ-001-05

description: "Alert feed filters work correctly"

test: "Select severity=CRITICAL → only CRITICAL alerts shown"

- criteria_id: AC-DVZ-001-06

description: "Dashboard is responsive on tablet (768px width)"

test: "Resize to 768px → layout adjusts, charts readable, no horizontal scroll"

FR-DVZ-002: Constituency Detail View

YAML

REQUIREMENT:

id: FR-DVZ-002

title: "Constituency-Level Detail Dashboard"

priority: MUST_HAVE

module: DASHBOARD

USER_STORY: >

AS an MP or District Authority,

I WANT to see detailed analytics for a specific constituency,

SO THAT I can understand the risk profile and take corrective action.

UI_SPECIFICATION:

page_route: "/dashboard/constituency/{constituency_id}"

layout:

header:

- Breadcrumb: "National > {State} > {Constituency}"
- MP name and photo placeholder
- Financial year selector

section_1_constituency_kpis:

cards:

- "Total Works Sanctioned"
- "Total Funds Released (₹ Cr)"
- "Fund Utilization Rate (%)"
- "Risk Score (with tier badge)"
- "Active Anomalies Count"

section_2_risk_breakdown:

type: "Radar/Spider chart"

title: "Risk Component Breakdown"

axes: ["Cost Overrun", "Delays", "Duplicates", "Patterns", "Fund Utilization"]

data: "Average component scores for constituency"

section_3_works_table:

type: "Sortable, filterable data table"

title: "All Works in Constituency"

columns:

- "Work ID (link)"
- "Description (truncated, expandable)"
- "Category"
- "Sanctioned Amount"
- "Actual Expenditure"
- "Cost Overrun %"
- "Status"
- "Delay (days)"
- "Risk Score (color badge)"
- "Anomaly Flags (icon list)"

features:

- "Sort by any column"

- "Filter by status, category, risk tier"
- "Search by work description"
- "Export to CSV"
- "Pagination: 25 per page"

section_4_expenditure_timeline:

type: "Stacked area chart"
 title: "Expenditure Over Time by Category"
 x_axis: "Month-Year"
 y_axis: "Cumulative expenditure (₹)"
 series: "One area per work_category"

section_5_duplicate_pairs:

type: "List/Card layout"
 title: "Potential Duplicate Works"
 visible_if: "duplicate anomalies exist for this constituency"
 each_card:

- "Work A description vs Work B description"
- "Similarity score"
- "Amount comparison"
- "Sanction date comparison"
- "Severity badge"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DVZ-002-01
 description: "Constituency page loads with correct data for selected constituency"
 test: "Navigate to constituency X → all KPIs match database aggregations for X"
- criteria_id: AC-DVZ-002-02
 description: "Works table supports sorting by risk score"
 test: "Click risk score column header → table sorted descending by risk_score"
- criteria_id: AC-DVZ-002-03
 description: "CSV export contains all visible table data"
 test: "Click export → downloaded CSV row count matches table total rows"
- criteria_id: AC-DVZ-002-04

description: "Duplicate pairs are displayed when they exist"

test: "Constituency with 2 duplicate pairs → both cards rendered with correct data"

FR-DVZ-003: Alert Management Interface

YAML

REQUIREMENT:

id: FR-DVZ-003

title: "Alert Management and Audit Trail"

priority: MUST_HAVE

module: DASHBOARD

USER STORY: >

AS a Ministry Official,

I WANT to view, acknowledge, and track all anomaly alerts with a full audit trail,

SO THAT I can ensure every flagged issue is reviewed and addressed.

UI_SPECIFICATION:

page_route: "/alerts"

features:

alert_list:

columns:

- "Alert ID"
- "Created At"
- "Anomaly Type"
- "Severity"
- "Constituency"
- "State"
- "Work ID"
- "Status: NEW | ACKNOWLEDGED | UNDER_REVIEW | RESOLVED | FALSE_POSITIVE"
- "Assigned To (dropdown of users)"

filters:

- "Status"
- "Severity"
- "Anomaly Type"
- "State"
- "Date Range"

bulk_actions:

- "Mark as Acknowledged"

- "Mark as False Positive"

alert_detail_panel:

on_click: "Slide-out panel from right"

content:

- "Full anomaly details"
- "Contextual data (work details, fund data)"
- "Risk score breakdown"
- "Status change history (audit trail)"
- "Notes/comments text area"
- "Action buttons: Acknowledge, Start Review, Resolve, Mark False Positive"

audit_trail:

stored_fields:

- "alert_id"
- "action (status_change, note_added, assignment_change)"
- "performed_by (user_id)"
- "timestamp"
- "old_value"
- "new_value"
- "note_text (if applicable)"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DVZ-003-01

description: "Alert status can be changed and change is logged"

test: "Change alert from NEW to ACKNOWLEDGED → audit_trail has entry with old=NEW, new=ACKNOWLEDGED, user_id, timestamp"

- criteria_id: AC-DVZ-003-02

description: "False positive marking excludes work from future anomaly counts in KPIs"

test: "Mark alert as FALSE_POSITIVE → KPI anomaly count decreases by 1"

- criteria_id: AC-DVZ-003-03

description: "Audit trail is immutable"

test: "Attempt to delete/modify audit_trail record via API → HTTP 403 or 405"

FR-DVZ-004: Report Generation

YAML

REQUIREMENT:

id: FR-DVZ-004

title: "Automated Report Generation"

priority: MUST_HAVE

module: DASHBOARD

USER_STORY: >

AS a Ministry Official,

I WANT to generate downloadable PDF and CSV reports summarizing

anomalies and risk assessments for any constituency, state, or national level,

SO THAT I can share findings with stakeholders who may not have system access.

INPUT_SPECIFICATION:

report_parameters:

scope: "NATIONAL | STATE | CONSTITUENCY"

scope_id: "string (state_name or constituency_id, required if scope != NATIONAL)"

financial_year: "string or ALL"

anomaly_types: "array (optional filter, default ALL)"

severity_filter: "array (optional, default ALL)"

output_format: "PDF | CSV"

OUTPUT_SPECIFICATION:

pdf_report_sections:

- "Cover page with report title, scope, generation date, generated by"
- "Executive summary: key metrics, total works, anomalies, risk distribution"
- "Anomaly summary table: count by type and severity"
- "Top 20 highest-risk works with details"
- "Duplicate work pairs (if any)"
- "Fund utilization analysis"
- "Trend charts (static images)"
- "Methodology note"

csv_report:

- "All works within scope with risk scores and anomaly flags"
- "One row per work, all columns from works_table view"

PROCESSING_RULES:

- "PDF generation using server-side rendering (e.g., WeasyPrint or ReportLab)"

- "Charts rendered as static PNG images embedded in PDF"
- "Report file stored temporarily (24 hours) for download"
- "Report generation logged in audit trail"
- "Maximum report generation time: 30 seconds"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-DVZ-004-01
description: "PDF report generated for constituency with correct data"
test: "Generate PDF for constituency X → PDF contains correct KPIs, top risk works match database"
- criteria_id: AC-DVZ-004-02
description: "CSV export contains all fields"
test: "Generate CSV → column count matches schema, row count matches scope filter"
- criteria_id: AC-DVZ-004-03
description: "Report generation completes within 30 seconds"
test: "Generate national report with 10K works → completion time < 30s"

3.5 Feature Module: Authentication & Authorization (AAA)

FR-AAA-001: User Authentication

YAML

REQUIREMENT:

id: FR-AAA-001
title: "JWT-based User Authentication"
priority: MUST_HAVE
module: AUTH

USER_STORY: >

AS any user,
I WANT to log in with credentials and receive a secure session,
SO THAT I can access the system according to my authorized role.

SPECIFICATION:

auth_method: "Username/password with JWT token"
token_details:
type: "JWT (RS256)"
access_token_expiry: "30 minutes"
refresh_token_expiry: "7 days"

payload_fields:

- "user_id"
- "role"
- "scope (state/constituency if applicable)"
- "iat (issued at)"
- "exp (expiry)"

endpoints:

- path: "POST /api/v1/auth/login"

input: { username: "string", password: "string" }

output_success: { access_token: "string", refresh_token: "string", user: { id, name, role } }

output_failure: { error: "INVALID_CREDENTIALS", status: 401 }

- path: "POST /api/v1/auth/refresh"

input: { refresh_token: "string" }

output_success: { access_token: "string" }

- path: "POST /api/v1/auth/logout"

action: "Invalidate refresh token"

password_policy:

min_length: 8

requires_uppercase: true

requires_lowercase: true

requires_digit: true

requires_special_char: true

hashing: "bcrypt with cost factor 12"

rate_limiting:

login_attempts: "5 per 15 minutes per IP"

lockout: "Account locked for 30 minutes after 5 failed attempts"

mvp_seed_users:

- { username: "admin", role: "ROLE_ADMIN", password: "Admin@1234" }

- { username: "ministry_user", role: "ROLE_MINISTRY", password: "Ministry@1234" }

- { username: "state_user", role: "ROLE_STATE_NODAL", scope: "Maharashtra", password: "State@1234" }

- { username: "district_user", role: "ROLE_DISTRICT", scope: "Pune", password: "District@1234" }

- { username: "mp_user", role: "ROLE_MP", scope: "Pune", password: "Mp@12345" }

ACCEPTANCE_CRITERIA:

- criteria_id: AC-AAA-001-01

description: "Valid credentials return JWT tokens"

test: "POST /auth/login with valid credentials → 200, body contains access_token and refresh_token"

- criteria_id: AC-AAA-001-02

description: "Invalid credentials return 401"

test: "POST /auth/login with wrong password → 401"

- criteria_id: AC-AAA-001-03

description: "Expired access token returns 401 on protected endpoint"

test: "Use expired JWT → GET /api/v1/dashboard → 401"

- criteria_id: AC-AAA-001-04

description: "Refresh token generates new access token"

test: "POST /auth/refresh with valid refresh_token → 200, new access_token"

- criteria_id: AC-AAA-001-05

description: "Account locks after 5 failed attempts"

test: "5 failed logins → 6th attempt returns 429 with lockout message"

FR-AAA-002: Role-Based Access Control

YAML

REQUIREMENT:

id: FR-AAA-002

title: "RBAC Enforcement on All Endpoints and Views"

priority: MUST_HAVE

module: AUTH

SPECIFICATION:

permission_matrix:

ROLE_ADMIN:

data_scope: "ALL"

can_upload_data: true

can_manage_users: true

can_view_all_dashboards: true

can_manage_alerts: true
can_generate_reports: true
can_configure_thresholds: true

ROLE_MINISTRY:

data_scope: "ALL"
can_upload_data: false
can_manage_users: false
can_view_all_dashboards: true
can_manage_alerts: true
can_generate_reports: true
can_configure_thresholds: false

ROLE_STATE_NODAL:

data_scope: "FILTER BY scope.state"
can_upload_data: false
can_manage_users: false
can_view_all_dashboards: false
can_view_state_dashboard: true
can_manage_alerts: "own state only"
can_generate_reports: "own state only"
can_configure_thresholds: false

ROLE_DISTRICT:

data_scope: "FILTER BY scope.district"
can_upload_data: false
can_manage_users: false
can_view_district_dashboard: true
can_manage_alerts: "own district only"
can_generate_reports: "own district only"

ROLE_MP:

data_scope: "FILTER BY scope.constituency"
can_upload_data: false
can_manage_users: false
can_view_constituency_dashboard: true
can_manage_alerts: false

can_generate_reports: "own constituency only"

ROLE_PUBLIC:

data_scope: "AGGREGATE ONLY (no work-level detail)"

can_upload_data: false

can_manage_users: false

can_view_public_dashboard: true

can_manage_alerts: false

can_generate_reports: false

enforcement:

- "Every API endpoint must check JWT role and scope before returning data"
- "Data queries must append WHERE clauses based on user scope"
- "UI must hide/disable features based on role permissions"
- "Unauthorized access attempts return HTTP 403"

ACCEPTANCE_CRITERIA:

- criteria_id: AC-AAA-002-01

description: "State nodal user can only see their state's data"

test: "Login as state_user (Maharashtra) → GET /api/v1/constituencies → only Maharashtra constituencies returned"

- criteria_id: AC-AAA-002-02

description: "MP user cannot access another constituency"

test: "Login as mp_user (Pune) → GET /api/v1/constituency/Mumbai → 403"

- criteria_id: AC-AAA-002-03

description: "Admin can access all data"

test: "Login as admin → GET /api/v1/constituencies → all constituencies returned"

- criteria_id: AC-AAA-002-04

description: "Public user cannot see work-level details"

test: "Login as public user → GET /api/v1/works → 403"

3.6 Feature Module: API Layer (API)

FR-API-001: RESTful API Endpoints

YAML

REQUIREMENT:

id: FR-API-001

title: "RESTful API for All Data Operations"

priority: MUST_HAVE

module: API

SPECIFICATION:

base_url: "/api/v1"

content_type: "application/json"

authentication: "Bearer JWT token in Authorization header"

endpoints:

Data Endpoints

- method: GET

path: "/works"

description: "List works with pagination, filtering, sorting"

query_params:

- { name: "page", type: "int", default: 1 }

- { name: "per_page", type: "int", default: 25, max: 100 }

- { name: "constituency", type: "string", optional: true }

- { name: "state", type: "string", optional: true }

- { name: "financial_year", type: "string", optional: true }

- { name: "status", type: "string", optional: true }

- { name: "risk_tier", type: "string", optional: true }

- { name: "sort_by", type: "string", default: "risk_score", options: ["risk_score", "sanctioned_amount", "sanction_date", "cost_overrun_percentage"] }

- { name: "sort_order", type: "string", default: "desc", options: ["asc", "desc"] }

response:

status: 200

body:

data: "array of work objects"

pagination: { page, per_page, total_records, total_pages }

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL", "ROLE_DISTRICT", "ROLE_MP"]

- method: GET

path: "/works/{work_id}"

description: "Get single work detail with risk breakdown and anomalies"

response:

body:

work: "full work object"

risk_score: "risk score object with component breakdown"

anomalies: "array of anomaly records for this work"

audit_trail: "array of status changes"

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL", "ROLE_DISTRICT", "ROLE_MP"]

- method: GET

path: "/constituencies"

description: "List constituencies with risk scores"

response:

body:

data: "array of constituency risk summary objects"

pagination: { page, per_page, total_records, total_pages }

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL"]

- method: GET

path: "/constituencies/{constituency_id}"

description: "Constituency detail with all analytics"

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL", "ROLE_DISTRICT", "ROLE_MP"]

Anomaly Endpoints

- method: GET

path: "/anomalies"

description: "List all detected anomalies"

query_params:

- { name: "type", type: "string", optional: true }

- { name: "severity", type: "string", optional: true }

- { name: "status", type: "string", optional: true }

- { name: "constituency", type: "string", optional: true }

- { name: "state", type: "string", optional: true }

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL", "ROLE_DISTRICT"]

- method: PATCH

path: "/anomalies/{anomaly_id}"

description: "Update anomaly status"

body: { status: "string", note: "string (optional)" }

roles: ["ROLE_ADMIN", "ROLE_MINISTRY"]

Analytics Endpoints

- method: GET

path: "/analytics/national-summary"

description: "National KPIs and summary statistics"

roles: ["ROLE_ADMIN", "ROLE_MINISTRY"]

- method: GET

path: "/analytics/state-summary/{state_name}"

description: "State-level summary"

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL"]

- method: GET

path: "/analytics/trends"

description: "Time series data for trend charts"

query_params:

- { name: "metric", type: "string", options: ["anomaly_count", "avg_risk_score", "fund_utilization", "expenditure"] }

- { name: "group_by", type: "string", options: ["financial_year", "month", "quarter"] }

- { name: "scope", type: "string", optional: true }

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL"]

Admin Endpoints

- method: POST

path: "/admin/upload"

description: "Upload data file"

content_type: "multipart/form-data"

roles: ["ROLE_ADMIN"]

- method: POST

path: "/admin/generate-synthetic"

description: "Trigger synthetic data generation"

body: { num_constituencies: "int", num_works_per_constituency: "int", anomaly_rate: "float" }

roles: ["ROLE_ADMIN"]

- method: POST

path: "/admin/run-detection"

description: "Trigger anomaly detection pipeline"

roles: ["ROLE_ADMIN"]

Report Endpoints

- method: POST

path: "/reports/generate"

description: "Generate report"

body: { scope: "string", scope_id: "string", financial_year: "string", format: "PDF|CSV" }

response: { report_id: "string", download_url: "string" }

roles: ["ROLE_ADMIN", "ROLE_MINISTRY", "ROLE_STATE_NODAL", "ROLE_DISTRICT", "ROLE_MP"]

- method: GET

path: "/reports/{report_id}/download"

description: "Download generated report"

response: "Binary file (PDF or CSV)"

error_handling:

standard_error_format:

body:

error:

code: "string (machine-readable)"

message: "string (human-readable)"

details: "object (optional additional context)"

error_codes:

- { code: "AUTH_REQUIRED", status: 401, message: "Authentication required" }

- { code: "FORBIDDEN", status: 403, message: "Insufficient permissions" }

- { code: "NOT_FOUND", status: 404, message: "Resource not found" }

- { code: "VALIDATION_ERROR", status: 422, message: "Input validation failed" }

- { code: "RATE_LIMITED", status: 429, message: "Too many requests" }

- { code: "INTERNAL_ERROR", status: 500, message: "Internal server error" }

ACCEPTANCE_CRITERIA:

- criteria_id: AC-API-001-01

description: "All listed endpoints are accessible and return correct status codes"

test: "Automated endpoint test suite hitting all endpoints with valid auth → all return expected status"

- criteria_id: AC-API-001-02

description: "Pagination works correctly"

test: "GET /works?page=2&per_page=10 with 25 total → returns 10 records, pagination shows total_pages=3"

- criteria_id: AC-API-001-03

description: "Filtering works correctly"

test: "GET /works?state=Maharashtra → all returned works have state=Maharashtra"

- criteria_id: AC-API-001-04

description: "Unauthorized endpoint access returns 401"

test: "GET /works without Authorization header → 401"

- criteria_id: AC-API-001-05

description: "OpenAPI 3.0 spec is auto-generated and accessible at /api/v1/docs"

test: "GET /api/v1/docs → returns valid Swagger UI page"

SECTION 4: NON-FUNCTIONAL REQUIREMENTS

4.1 Performance Requirements

YAML

PERFORMANCE:

- id: NFR-PERF-001

requirement: "Dashboard page load time"

metric: "Time to Interactive (TTI)"

target: "< 3 seconds on 10 Mbps connection"

measurement: "Lighthouse performance audit"

test_data_volume: "10,000 work records, 50 constituencies"

- id: NFR-PERF-002

requirement: "API response time"

metric: "95th percentile latency"

target:

list_endpoints: "< 500ms"

detail_endpoints: "< 300ms"

analytics_endpoints: "< 1000ms"

measurement: "Load testing with 50 concurrent users"

- id: NFR-PERF-003

requirement: "Anomaly detection pipeline execution"

metric: "Full pipeline completion time"

target: "< 120 seconds for 10,000 work records"

measurement: "End-to-end pipeline timing from trigger to completion"

- id: NFR-PERF-004

requirement: "Data upload processing"

metric: "CSV ingestion throughput"

target: "1000 records/second"

measurement: "Upload 10,000 row CSV, measure ingestion time"

- id: NFR-PERF-005

requirement: "Concurrent user support"

metric: "Simultaneous active sessions"

target: "50 concurrent users without degradation"

measurement: "Load test with 50 simulated users performing typical workflows"

- id: NFR-PERF-006

requirement: "Database query performance"

metric: "95th percentile query latency"

target: "< 200ms for indexed queries"

measurement: "pg_stat_statements monitoring"

- id: NFR-PERF-007

requirement: "Report generation time"

metric: "PDF report creation latency"

target: "< 30 seconds for constituency-level report"

measurement: "Timer on report generation endpoint"

4.2 Security Requirements

YAML

SECURITY:

- id: NFR-SEC-001

requirement: "Data encryption at rest"

standard: "AES-256"

scope: "All database files and uploaded data files"

implementation: "PostgreSQL TDE or filesystem-level encryption"

- id: NFR-SEC-002

requirement: "Data encryption in transit"

standard: "TLS 1.3 minimum"

scope: "All HTTP communications"

implementation: "HTTPS with valid SSL certificate; HSTS header"

- id: NFR-SEC-003

requirement: "Authentication security"

standard: "OWASP Authentication Guidelines"

implementation:

- "bcrypt password hashing (cost factor 12)"
- "JWT with RS256 signing"
- "HttpOnly, Secure, SameSite=Strict cookies for refresh tokens"
- "CSRF protection on state-changing endpoints"

- id: NFR-SEC-004

requirement: "Input validation and injection prevention"

standard: "OWASP Top 10"

implementation:

- "Parameterized SQL queries (no string concatenation)"
- "Input sanitization on all user-supplied text"
- "XSS prevention with Content-Security-Policy header"
- "File upload type validation (MIME type + extension)"

- id: NFR-SEC-005

requirement: "Audit logging"

standard: "Government cybersecurity guidelines"

implementation:

- "All authentication events logged (login, logout, failed attempts)"
- "All data modification events logged (upload, anomaly status change)"
- "All report generation events logged"
- "Logs include: timestamp, user_id, IP, action, resource, result"
- "Log retention: 1 year minimum"
- "Logs stored in append-only format (tamper-resistant)"

- id: NFR-SEC-006

requirement: "API security"

implementation:

- "Rate limiting: 100 requests/minute per user"

- "Request size limit: 50MB"
- "CORS restricted to allowed origins"
- "API keys for service-to-service communication"

- id: NFR-SEC-007

requirement: "Session management"

implementation:

- "Access token expiry: 30 minutes"
- "Refresh token expiry: 7 days"
- "Token rotation on refresh"
- "Immediate invalidation on logout"
- "Maximum 3 concurrent sessions per user"

- id: NFR-SEC-008

requirement: "Dependency security"

implementation:

- "No known CVE vulnerabilities in production dependencies"
- "Automated dependency vulnerability scanning (e.g., pip-audit, npm audit)"
- "Docker base images from trusted sources only"

4.3 Accessibility Requirements

YAML

ACCESSIBILITY:

- id: NFR-ACC-001

standard: "WCAG 2.1 Level AA"

requirements:

perceivable:

- "All images and charts have meaningful alt text"
- "Color is not the sole means of conveying information"
- "Minimum contrast ratio: 4.5:1 for normal text, 3:1 for large text"
- "Text resizable to 200% without loss of content"
- "All form inputs have associated labels"

operable:

- "All interactive elements are keyboard accessible"
- "Visible focus indicators on all focusable elements"
- "No keyboard traps"
- "Skip navigation link provided"

- "No content that flashes more than 3 times per second"

understandable:

- "Page language declared in HTML lang attribute (en)"
- "Form validation errors clearly described and linked to fields"
- "Consistent navigation across pages"
- "Error messages in plain language with correction guidance"

robust:

- "Valid HTML5 markup"
- "ARIA landmarks for main regions (header, nav, main, footer)"
- "ARIA roles and properties on custom widgets"
- "Compatible with screen readers (NVDA, JAWS, VoiceOver)"

testing:

- "Automated: axe-core accessibility engine integrated in test suite"
- "Target: 0 critical/serious axe violations"
- "Keyboard navigation test: all features accessible via keyboard only"

- id: NFR-ACC-002

standard: "GIGW 3.0 Compliance"

requirements:

- "Government of India emblem/logo placement (if applicable)"
- "bilingual support placeholder (Hindi/English — English only for MVP)"
- "Sitemap page"
- "Terms of use and privacy policy pages"
- "Last updated timestamp on all pages"
- "Website policies page (accessibility statement, copyright, privacy)"

4.4 Reliability & Availability

YAML

RELIABILITY:

- id: NFR-REL-001

requirement: "System uptime"

target: "99% uptime during demonstration period"

measurement: "Uptime monitoring (health check endpoint)"

- id: NFR-REL-002

requirement: "Error rate"

target: "< 1% of API requests result in 5xx errors"

measurement: "Error rate monitoring from access logs"

- id: NFR-REL-003

requirement: "Data integrity"

target: "Zero data corruption or loss during normal operation"

implementation:

- "Database transaction isolation level: READ COMMITTED minimum"
- "Foreign key constraints enforced"
- "Application-level data validation before persistence"

- id: NFR-REL-004

requirement: "Graceful degradation"

target: "Dashboard remains viewable with cached data if analytics engine is temporarily unavailable"

implementation: "Last-computed analytics cached; stale indicator shown if > 1 hour old"

- id: NFR-REL-005

requirement: "Backup and recovery"

target: "Database backup every 6 hours; recovery within 1 hour"

implementation: "pg_dump scheduled backup; documented restoration procedure"

- id: NFR-REL-006

requirement: "Health monitoring"

implementation:

- "GET /health endpoint returning system status"
- "Response includes: database_connected, ml_engine_ready, disk_space_ok, last_detection_run"
- "Response time < 100ms"

4.5 Scalability

YAML

SCALABILITY:

- id: NFR-SCL-001

requirement: "Data volume"

current_target: "50,000 work records (MVP)"

future_target: "5,000,000 work records (post-MVP)"

design_implications:

- "Database indexed on frequently queried columns"

- "Pagination on all list endpoints"
- "Asynchronous processing for heavy computations"

- id: NFR-SCL-002

requirement: "Horizontal scalability"

design: "Stateless API servers behind load balancer"

implementation: "Docker containerized, ready for Kubernetes deployment post-MVP"

4.6 Maintainability

YAML

MAINTAINABILITY:

- id: NFR-MNT-001

requirement: "Code quality"

standards:

python: "PEP 8, type hints on all functions, docstrings on all public methods"

javascript: "ESLint with recommended config, PropTypes/TypeScript interfaces"

targets:

- "No linting errors in CI"
- "Cyclomatic complexity per function: < 15"

- id: NFR-MNT-002

requirement: "Test coverage"

target: "Minimum 70% line coverage on backend code"

measurement: "pytest-cov for Python, Jest coverage for frontend"

- id: NFR-MNT-003

requirement: "Documentation"

deliverables:

- "README.md with setup instructions"
- "API documentation via Swagger/OpenAPI"
- "Database schema documentation (ERD)"
- "Anomaly detection methodology document"
- "Deployment guide"

- id: NFR-MNT-004

requirement: "Logging"

implementation:

- "Structured JSON logging"

- "Log levels: DEBUG, INFO, WARNING, ERROR, CRITICAL"
 - "Request ID tracking across log entries"
 - "No sensitive data (passwords, tokens) in logs"
-

SECTION 5: SYSTEM DESIGN CONSTRAINTS

5.1 Technology Stack (Mandatory)

YAML

TECHNOLOGY_STACK:

backend:

language: "Python 3.11+"

framework: "FastAPI 0.100+"

justification: "High performance async framework, auto-generated OpenAPI docs, type validation"

alternatives_considered: "Django (heavier), Flask (less structured)"

frontend:

language: "JavaScript/TypeScript"

framework: "React 18+ with Vite build tool"

ui_library: "Ant Design or Material UI (government-style, accessible components)"

charting: "Recharts or Apache ECharts"

mapping: "Leaflet.js with India GeoJSON"

state_management: "React Context API + React Query (TanStack Query)"

database:

primary: "PostgreSQL 15+"

extensions:

- "pgvector (for text embedding similarity search)"

orm: "SQLAlchemy 2.0+ with Alembic migrations"

ml_libraries:

- "scikit-learn (Isolation Forest, statistical analysis)"
- "sentence-transformers (all-MiniLM-L6-v2 for text embeddings)"
- "pandas (data manipulation)"
- "numpy (numerical computation)"

infrastructure:

containerization: "Docker with docker-compose"

reverse_proxy: "Nginx (HTTPS termination, static file serving)"

task_queue: "Celery with Redis broker (for async ML pipeline execution)"

cache: "Redis 7+"

testing:

backend: "pytest, pytest-asyncio, httpx (async test client)"

frontend: "Jest, React Testing Library"

e2e: "Playwright (optional for MVP)"

api: "pytest + OpenAPI schema validation"

documentation:

api: "Auto-generated Swagger/OpenAPI via FastAPI"

code: "Inline docstrings + README"

version_control:

system: "Git"

branching: "main (production), develop (integration)"

5.2 Deployment Environment

YAML

DEPLOYMENT:

mvp_environment:

type: "Docker Compose on single server"

minimum_specs:

cpu: "4 vCPU"

ram: "8 GB"

storage: "50 GB SSD"

os: "Ubuntu 22.04 LTS"

docker_services:

- service: "api"

image: "python:3.11-slim"

ports: "8000"

resources: "2 vCPU, 3 GB RAM"

- service: "frontend"

image: "node:20-alpine (build) → nginx:alpine (serve)"

ports: "3000 → 80"

resources: "0.5 vCPU, 512 MB RAM"

- service: "postgres"

image: "pgvector/pgvector:pg16"

ports: "5432 (internal only)"

volumes: "persistent data volume"

resources: "1 vCPU, 2 GB RAM"

- service: "redis"

image: "redis:7-alpine"

ports: "6379 (internal only)"

resources: "0.5 vCPU, 512 MB RAM"

- service: "celery_worker"

image: "same as api"

resources: "1 vCPU, 2 GB RAM"

- service: "nginx"

image: "nginx:alpine"

ports: "80, 443"

config: "reverse proxy to api and frontend"

production_readiness_notes:

- "Docker Compose suitable for MVP demo"
- "Production deployment should use Kubernetes on NIC/Government cloud"
- "Database should use managed PostgreSQL service"
- "SSL certificates from NIC CA or government-approved CA"

environment_variables:

required:

- "DATABASE_URL: PostgreSQL connection string"
- "REDIS_URL: Redis connection string"
- "JWT_SECRET_KEY: RSA private key path"
- "JWT_PUBLIC_KEY: RSA public key path"
- "CORS_ORIGINS: comma-separated allowed origins"
- "LOG_LEVEL: DEBUG|INFO|WARNING|ERROR"
- "ENVIRONMENT: development|staging|production"

optional:

- "SENTRY_DSN: error tracking URL"
- "ML_MODEL_PATH: directory for trained model artifacts"

5.3 Data Privacy & Regulatory Constraints

YAML

DATA_PRIVACY:

applicable_regulations:

- name: "Digital Personal Data Protection Act 2023 (DPDPA)"
 - applicability: "If system processes any personal data of MPs, officers, or citizens"
 - mvp_compliance:
 - "Minimize personal data collection to operational necessity"
 - "MP names and constituency data are public information — exempt from consent"
 - "No citizen personal data processed in MVP"
 - "Implement data access logging"

- name: "IT Act 2000 / IT Amendment Act 2008"
 - applicability: "All government IT systems"
 - mvp_compliance:
 - "Reasonable security practices (encryption, access control)"
 - "Data integrity protections"

- name: "Government of India IT Security Policy"
 - applicability: "All government applications"
 - mvp_compliance:
 - "Role-based access control"
 - "Audit logging"
 - "No data transmission to external services"
 - "All processing on government-controlled infrastructure"

data_classification:

public_data:

- "Aggregate fund statistics (already on public dashboard)"
- "Constituency-level summaries"

description: "Can be displayed to ROLE_PUBLIC"

internal_data:

- "Individual work details with risk scores"
- "Anomaly flags and investigations"

- "Alert audit trails"

description: "Restricted to authenticated government users per RBAC"

sensitive_data:

- "User credentials (hashed)"

- "System configuration"

- "ML model internals"

description: "Admin access only"

constraints:

- "ALL data processing must occur within the deployment infrastructure — no external API calls for data processing"

- "ML models must run locally (no cloud ML services like OpenAI, Google ML, etc.)"

- "Sentence transformer model must be downloaded and bundled in container"

- "No analytics or telemetry data sent to external services"

- "Database must not be accessible from public internet"

5.4 Integration Constraints

YAML

INTEGRATION_CONSTRAINTS:

existing_system_integration:

mplads_portal:

integration_type: "Read-only data consumption"

method: "CSV export/import or web scraping of public dashboard"

constraint: "No write access to MPLADS production database"

constraint: "No API available — data must be manually exported or scraped"

external_service_restrictions:

- "NO external AI/ML API calls (OpenAI, Claude, Google AI, etc.)"

- "NO external database services"

- "NO cloud storage services (S3, GCS, etc.)"

- "ALL dependencies must be self-hosted within Docker deployment"

- "NLP model must be open-source and locally deployable"

browser_compatibility:

minimum: "Chrome 90+, Firefox 88+, Edge 90+, Safari 14+"

testing: "All features verified on Chrome latest"

network_assumptions:

- "System deployed on internal government network or secure VPN"
- "Users access via modern web browsers"
- "Minimum bandwidth: 10 Mbps"

5.5 Database Schema

YAML

DATABASE_SCHEMA:

tables:

- table: "constituencies"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "name", type: "VARCHAR(255)", not_null: true, unique: true }
- { name: "state", type: "VARCHAR(100)", not_null: true }
- { name: "mp_name", type: "VARCHAR(255)" }
- { name: "mp_type", type: "VARCHAR(20)", enum: ["LOK_SABHA", "RAJYA_SABHA"] }
- { name: "created_at", type: "TIMESTAMP", default: "NOW()" }
- { name: "updated_at", type: "TIMESTAMP", default: "NOW()" }

indexes:

- { columns: ["state"], type: "btree" }
- { columns: ["name"], type: "btree" }

- table: "works"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "work_id", type: "VARCHAR(50)", not_null: true, unique: true }
- { name: "constituency_id", type: "UUID", fk: "constituencies.id", not_null: true }
- { name: "work_description", type: "TEXT", not_null: true }
- { name: "work_category", type: "VARCHAR(50)", not_null: true }
- { name: "sanctioned_amount", type: "DECIMAL(15,2)", not_null: true }
- { name: "actual_expenditure", type: "DECIMAL(15,2)", default: 0 }
- { name: "cost_overrun_percentage", type: "DECIMAL(8,2)", computed: true }
- { name: "sanction_date", type: "DATE", not_null: true }
- { name: "expected_completion_date", type: "DATE" }
- { name: "completion_date", type: "DATE" }
- { name: "work_status", type: "VARCHAR(20)", not_null: true }
- { name: "implementing_agency", type: "VARCHAR(255)" }
- { name: "financial_year", type: "VARCHAR(10)", not_null: true }
- { name: "latitude", type: "DECIMAL(10,7)" }

- { name: "longitude", type: "DECIMAL(10,7)" }
- { name: "description_embedding", type: "VECTOR(384)" }
- { name: "risk_score", type: "INTEGER", default: 0 }
- { name: "risk_tier", type: "VARCHAR(10)", default: "LOW" }
- { name: "created_at", type: "TIMESTAMP", default: "NOW()" }
- { name: "updated_at", type: "TIMESTAMP", default: "NOW()" }

indexes:

- { columns: ["constituency_id"], type: "btree" }
- { columns: ["financial_year"], type: "btree" }
- { columns: ["work_status"], type: "btree" }
- { columns: ["risk_score"], type: "btree" }
- { columns: ["work_category"], type: "btree" }
- { columns: ["description_embedding"], type: "ivfflat", ops: "vector_cosine_ops" }

- table: "fund_releases"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "release_id", type: "VARCHAR(50)", not_null: true, unique: true }
- { name: "constituency_id", type: "UUID", fk: "constituencies.id", not_null: true }
- { name: "financial_year", type: "VARCHAR(10)", not_null: true }
- { name: "installment_number", type: "INTEGER", not_null: true }
- { name: "amount_released", type: "DECIMAL(15,2)", not_null: true }
- { name: "release_date", type: "DATE", not_null: true }
- { name: "cumulative_release", type: "DECIMAL(15,2)" }
- { name: "created_at", type: "TIMESTAMP", default: "NOW()" }

indexes:

- { columns: ["constituency_id", "financial_year"], type: "btree" }

- table: "anomalies"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "work_id", type: "UUID", fk: "works.id", nullable: true }
- { name: "constituency_id", type: "UUID", fk: "constituencies.id", not_null: true }
- { name: "anomaly_type", type: "VARCHAR(50)", not_null: true }
- { name: "severity", type: "VARCHAR(10)", not_null: true }
- { name: "confidence_score", type: "DECIMAL(5,4)" }
- { name: "detection_method", type: "VARCHAR(50)" }

- { name: "details", type: "JSONB" }
- { name: "status", type: "VARCHAR(20)", default: "NEW" }
- { name: "detected_at", type: "TIMESTAMP", default: "NOW()" }
- { name: "updated_at", type: "TIMESTAMP", default: "NOW()" }

indexes:

- { columns: ["anomaly_type"], type: "btree" }
- { columns: ["severity"], type: "btree" }
- { columns: ["status"], type: "btree" }
- { columns: ["constituency_id"], type: "btree" }
- { columns: ["detected_at"], type: "btree" }

- table: "duplicate_pairs"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "work_id_a", type: "UUID", fk: "works.id", not_null: true }
- { name: "work_id_b", type: "UUID", fk: "works.id", not_null: true }
- { name: "text_similarity", type: "DECIMAL(5,4)", not_null: true }
- { name: "amount_similarity", type: "DECIMAL(5,4)" }
- { name: "composite_score", type: "INTEGER" }
- { name: "anomaly_id", type: "UUID", fk: "anomalies.id" }
- { name: "detected_at", type: "TIMESTAMP", default: "NOW()" }

constraints:

- "UNIQUE(work_id_a, work_id_b)"
- "CHECK(work_id_a < work_id_b) -- prevent reverse duplicates"

- table: "constituency_risk_scores"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "constituency_id", type: "UUID", fk: "constituencies.id", not_null: true }
- { name: "financial_year", type: "VARCHAR(10)", not_null: true }
- { name: "risk_score", type: "INTEGER", not_null: true }
- { name: "risk_tier", type: "VARCHAR(10)", not_null: true }
- { name: "total_works", type: "INTEGER" }
- { name: "high_risk_works", type: "INTEGER" }
- { name: "fund_utilization_rate", type: "DECIMAL(8,2)" }
- { name: "total_funds_released", type: "DECIMAL(15,2)" }
- { name: "total_expenditure", type: "DECIMAL(15,2)" }

- { name: "calculated_at", type: "TIMESTAMP", default: "NOW()" }

indexes:

- { columns: ["constituency_id", "financial_year"], type: "btree", unique: true }

- { columns: ["risk_score"], type: "btree" }

- table: "users"

columns:

- { name: "id", type: "UUID", pk: true }

- { name: "username", type: "VARCHAR(100)", not_null: true, unique: true }

- { name: "password_hash", type: "VARCHAR(255)", not_null: true }

- { name: "full_name", type: "VARCHAR(255)" }

- { name: "role", type: "VARCHAR(30)", not_null: true }

- { name: "scope_type", type: "VARCHAR(20)", nullable: true, enum: ["STATE", "DISTRICT", "CONSTITUENCY"] }

- { name: "scope_value", type: "VARCHAR(255)", nullable: true }

- { name: "is_active", type: "BOOLEAN", default: true }

- { name: "failed_login_attempts", type: "INTEGER", default: 0 }

- { name: "locked_until", type: "TIMESTAMP", nullable: true }

- { name: "created_at", type: "TIMESTAMP", default: "NOW()" }

- { name: "last_login", type: "TIMESTAMP", nullable: true }

- table: "audit_log"

columns:

- { name: "id", type: "UUID", pk: true }

- { name: "user_id", type: "UUID", fk: "users.id" }

- { name: "action", type: "VARCHAR(50)", not_null: true }

- { name: "resource_type", type: "VARCHAR(50)" }

- { name: "resource_id", type: "VARCHAR(255)" }

- { name: "old_value", type: "JSONB" }

- { name: "new_value", type: "JSONB" }

- { name: "ip_address", type: "INET" }

- { name: "user_agent", type: "TEXT" }

- { name: "timestamp", type: "TIMESTAMP", default: "NOW()" }

indexes:

- { columns: ["user_id"], type: "btree" }

- { columns: ["action"], type: "btree" }

- { columns: ["timestamp"], type: "btree" }

constraints:

- "Table is INSERT-only; no UPDATE or DELETE allowed (enforced by application and DB trigger)"

- table: "upload_history"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "user_id", type: "UUID", fk: "users.id", not_null: true }
- { name: "filename", type: "VARCHAR(255)", not_null: true }
- { name: "file_hash", type: "VARCHAR(64)", not_null: true }
- { name: "file_size_bytes", type: "INTEGER" }
- { name: "records_total", type: "INTEGER" }
- { name: "records_valid", type: "INTEGER" }
- { name: "records_rejected", type: "INTEGER" }
- { name: "validation_errors", type: "JSONB" }
- { name: "status", type: "VARCHAR(20)", enum: ["PROCESSING", "COMPLETED", "FAILED"] }
- { name: "uploaded_at", type: "TIMESTAMP", default: "NOW()" }
- { name: "completed_at", type: "TIMESTAMP" }

- table: "detection_runs"

columns:

- { name: "id", type: "UUID", pk: true }
- { name: "triggered_by", type: "UUID", fk: "users.id" }
- { name: "trigger_type", type: "VARCHAR(20)", enum: ["MANUAL", "AUTO_POST_UPLOAD"] }
- { name: "status", type: "VARCHAR(20)", enum: ["RUNNING", "COMPLETED", "FAILED"] }
- { name: "anomalies_detected", type: "INTEGER" }
- { name: "works_analyzed", type: "INTEGER" }
- { name: "started_at", type: "TIMESTAMP", default: "NOW()" }
- { name: "completed_at", type: "TIMESTAMP" }
- { name: "error_message", type: "TEXT" }

SECTION 6: IMPLEMENTATION ROADMAP

6.1 Phased MVP Development Timeline

YAML

ROADMAP:

total_duration: "36 hours (solo hackathon)"

methodology: "Iterative build with vertical slices"

PHASE_1:

name: "Foundation & Data Layer"

duration: "Hours 0-8"

deliverables:

hour_0_to_2:

- "Project scaffolding (FastAPI backend, React frontend, Docker Compose)"
- "PostgreSQL + pgvector database setup"
- "Database schema creation via Alembic migrations"
- "Environment configuration (.env, docker-compose.yml)"
- "Basic health check endpoint"

checkpoint: "Docker Compose up → all services healthy → /health returns 200"

hour_2_to_4:

- "Synthetic data generator implementation (FR-DIM-002)"
- "Generate 50 constituencies × 100 works = 5000 records"
- "Include ground-truth anomaly labels"
- "Data validation utility functions"

checkpoint: "Synthetic CSV files generated → pass schema validation → load into DB"

hour_4_to_6:

- "CSV upload endpoint (FR-DIM-001)"
- "Data ingestion pipeline with validation"
- "Upload history logging"
- "Database seed with synthetic data"

checkpoint: "Upload synthetic CSV via API → 5000 records in works table → upload_history entry created"

hour_6_to_8:

- "User authentication (FR-AAA-001)"
- "JWT token generation and validation"
- "RBAC middleware (FR-AAA-002)"
- "Seed users creation"
- "Login/logout endpoints"

checkpoint: "Login with admin credentials → receive JWT → access protected endpoint → 200"

PHASE_2:

name: "Anomaly Detection Engine"

duration: "Hours 8-18"

deliverables:

hour_8_to_10:

- "Cost overrun detection (FR-ADE-001)"
- "Threshold-based rules"
- "Z-score statistical detection"
- "Anomaly records creation in database"

checkpoint: "Run cost overrun detection → anomalies table populated → matches ground truth labels $\geq 70\%$ precision"

hour_10_to_12:

- "Delayed project detection (FR-ADE-003)"
- "Stalled project detection"
- "Fund utilization anomaly detection (FR-ADE-004)"

checkpoint: "Run delay + fund detection → additional anomalies in DB → correct severity assignments"

hour_12_to_15:

- "Duplicate work detection (FR-ADE-002)"
- "Text embedding generation using sentence-transformers"
- "pgvector similarity search"
- "Composite duplicate scoring"

checkpoint: "Run duplicate detection → duplicate_pairs table populated → known synthetic duplicates detected"

hour_15_to_17:

- "Suspicious pattern detection (FR-ADE-005)"
- "Amount clustering, end-of-year rush, round number bias"
- "Risk scoring engine (FR-RSE-001)"
- "Composite score calculation for all works and constituencies"

checkpoint: "All works have risk_score populated → constituency_risk_scores table populated → distribution matches expected tiers"

hour_17_to_18:

- "Celery task for async pipeline execution"
- "Pipeline orchestration: trigger all detectors in sequence"
- "Detection run logging"

checkpoint: "POST /admin/run-detection → Celery task executes → detection_runs table shows COMPLETED"

PHASE_3:

name: "API Layer & Dashboard"

duration: "Hours 18-30"

deliverables:

hour_18_to_21:

- "REST API endpoints for works, constituencies, anomalies (FR-API-001)"
- "Pagination, filtering, sorting"
- "RBAC enforcement on all endpoints"
- "OpenAPI documentation generation"

checkpoint: "All API endpoints return correct data with proper auth → Swagger docs accessible"

hour_21_to_24:

- "React frontend scaffolding"
- "Login page"
- "National overview dashboard (FR-DVZ-001)"
- "KPI cards"
- "India heatmap (Leaflet.js)"
- "Alert feed table"

checkpoint: "Login → dashboard loads → KPIs show correct numbers → map renders with colors"

hour_24_to_27:

- "Constituency detail page (FR-DVZ-002)"
- "Works data table with sorting/filtering"
- "Risk breakdown chart"
- "Expenditure timeline chart"
- "Duplicate pairs display"

checkpoint: "Click constituency → detail page loads → all sections render with correct data"

hour_27_to_29:

- "Alert management interface (FR-DVZ-003)"
- "Alert status workflow"
- "Audit trail display"
- "Bar chart (top risk constituencies)"
- "Donut chart (anomaly distribution)"
- "Trend line chart"

checkpoint: "Alert status change → audit trail entry created → UI updates"

hour_29_to_30:

- "Report generation (FR-DVZ-004)"
- "CSV export from works table"

- "PDF report generation (basic template)"

checkpoint: "Generate PDF report → downloads correctly → contains expected sections"

PHASE_4:

name: "Testing, Polish & Submission"

duration: "Hours 30-36"

deliverables:

hour_30_to_32:

- "Backend unit tests (pytest)"
- "API integration tests"
- "Anomaly detection validation against ground truth"
- "Fix critical bugs found during testing"

checkpoint: "pytest passes with $\geq 70\%$ coverage → all API tests pass"

hour_32_to_34:

- "Frontend polish: responsive layout, accessibility checks"
- "Error handling and loading states"
- "WCAG accessibility audit (axe-core)"
- "Cross-browser check (Chrome, Firefox)"

checkpoint: "axe-core reports 0 critical violations → responsive layout works on 768px"

hour_34_to_35:

- "End-to-end demo flow verification"
- "Documentation: README, setup guide, demo script"
- "Docker Compose cleanup and optimization"
- "Environment variable documentation"

checkpoint: "Fresh docker-compose up → follow demo script → all features work end-to-end"

hour_35_to_36:

- "Final demo recording (optional)"
- "Code cleanup and final commit"
- "Submission package preparation"

checkpoint: "Git repository clean → docker-compose up works from scratch → README complete"

6.2 Critical Path Dependencies

YAML

DEPENDENCIES:

- task: "Anomaly Detection"

depends_on: ["Database Schema", "Synthetic Data Generator", "Data Ingestion"]

- task: "Risk Scoring"

depends_on: ["All Anomaly Detectors"]

- task: "Dashboard"

depends_on: ["API Endpoints", "Anomaly Detection (for data to display)"]

- task: "Report Generation"

depends_on: ["API Endpoints", "Risk Scoring"]

- task: "Alert Management"

depends_on: ["Anomaly Detection", "API Endpoints"]

6.3 Risk Mitigation

YAML

RISKS:

- risk: "MPLADS dashboard scraping fails (site structure changes or blocking)"

probability: MEDIUM

impact: HIGH

mitigation: "Synthetic data generator serves as primary data source for MVP demo; scraping is SHOULD_HAVE, not MUST_HAVE"

- risk: "Sentence transformer model too large / slow for hackathon compute"

probability: LOW

impact: MEDIUM

mitigation: "Use all-MiniLM-L6-v2 (80MB, fast inference); fallback to TF-IDF + cosine similarity if needed"

- risk: "Time overrun on frontend development"

probability: MEDIUM

impact: MEDIUM

mitigation: "Use Ant Design pre-built components; prioritize national dashboard over detail views"

- risk: "Docker resource constraints"

probability: LOW

impact: HIGH

mitigation: "Optimize container images; use slim base images; limit concurrent services if needed"

SECTION 7: VERIFICATION & VALIDATION CRITERIA

7.1 Automated Test Suite

YAML

TEST_SUITE:

backend_unit_tests:

framework: "pytest"

location: "tests/"

categories:

- category: "Data Validation Tests"

test_file: "tests/test_data_validation.py"

tests:

- test_id: T-DV-001

name: "test_valid_csv_schema_accepted"

action: "Submit CSV with all valid columns and data types"

expected: "Validation returns no errors, all records accepted"

- test_id: T-DV-002

name: "test_missing_required_column_rejected"

action: "Submit CSV missing 'work_id' column"

expected: "Validation returns error specifying missing column"

- test_id: T-DV-003

name: "test_invalid_date_format_rejected"

action: "Submit CSV with sanction_date in DD/MM/YYYY format"

expected: "Row rejected with error specifying expected YYYY-MM-DD format"

- test_id: T-DV-004

name: "test_negative_amount_rejected"

action: "Submit CSV with sanctioned_amount = -5000"

expected: "Row rejected with constraint violation error"

- test_id: T-DV-005

name: "test_duplicate_work_id_handled"

action: "Upload two CSVs with overlapping work_ids"

expected: "Second upload updates existing records, audit log entry created"

- category: "Anomaly Detection Tests"

test_file: "tests/test_anomaly_detection.py"

tests:

- test_id: T-AD-001

name: "test_cost_overrun_threshold_detection"

action: "Insert work with sanctioned=100000, actual=125000"

expected: "Anomaly detected: type=COST_OVERRUN, severity=MEDIUM"

- test_id: T-AD-002

name: "test_cost_overrun_not_flagged_below_threshold"

action: "Insert work with sanctioned=100000, actual=110000"

expected: "No COST_OVERRUN anomaly created for this work"

- test_id: T-AD-003

name: "test_cost_overrun_critical_severity"

action: "Insert work with sanctioned=100000, actual=200000"

expected: "Anomaly detected: type=COST_OVERRUN, severity=CRITICAL"

- test_id: T-AD-004

name: "test_delay_detection_medium"

action: "Insert work with expected_completion 100 days ago, status=IN_PROGRESS"

expected: "Anomaly detected: type=DELAYED_PROJECT, severity=MEDIUM, delay_days=100"

- test_id: T-AD-005

name: "test_delay_detection_critical"

action: "Insert work with expected_completion 400 days ago, status=IN_PROGRESS"

expected: "Anomaly detected: type=DELAYED_PROJECT, severity=CRITICAL"

- test_id: T-AD-006

name: "test_completed_work_not_flagged_as_delayed"

action: "Insert work completed before expected_completion_date"

expected: "No DELAYED_PROJECT anomaly"

- test_id: T-AD-007

name: "test_stalled_project_detection"

action: "Insert work sanctioned 400 days ago, status=SANCTIONED"

expected: "Anomaly detected: type=STALLED_PROJECT, severity=HIGH"

- test_id: T-AD-008

name: "test_duplicate_detection_similar_descriptions"

action: |

Insert two works in same constituency:

A: "Construction of community hall at Village Rampur"

B: "Construction of community hall in Village Rampur"

Similar amounts, dates within 6 months

expected: "Duplicate pair detected with text_similarity > 0.85"

- test_id: T-AD-009

name: "test_duplicate_not_flagged_different_descriptions"

action: |

Insert two works in same constituency:

A: "Construction of primary school building"

B: "Installation of solar street lights in market area"

expected: "No duplicate pair created"

- test_id: T-AD-010

name: "test_low_fund_utilization_detection"

action: "Create constituency with released=5000000, expenditure=1000000 (20%)"

expected: "Anomaly detected: type=LOW_UTILIZATION, severity=HIGH"

- test_id: T-AD-011

name: "test_normal_fund_utilization_not_flagged"

action: "Create constituency with released=5000000, expenditure=4000000 (80%)"

expected: "No fund utilization anomaly"

- test_id: T-AD-012

name: "test_amount_clustering_detection"

action: "Insert 7 works in same constituency with identical sanctioned_amount=500000"

expected: "Anomaly detected: type=AMOUNT_CLUSTERING"

- category: "Risk Scoring Tests"

test_file: "tests/test_risk_scoring.py"

tests:

- test_id: T-RS-001

name: "test_zero_risk_score_for_clean_work"

action: "Work with no anomalies"
expected: "risk_score=0, risk_tier=LOW"

- test_id: T-RS-002

name: "test_high_risk_score_for_multiple_anomalies"
action: "Work with 40% cost overrun + 200 day delay + possible duplicate"
expected: "risk_score >= 50, risk_tier IN [HIGH, CRITICAL]"

- test_id: T-RS-003

name: "test_constituency_score_aggregation"
action: "Constituency with 10 works, 5 having score > 50"
expected: "constituency_risk_score reflects high proportion of risky works"

- category: "Authentication Tests"

test_file: "tests/test_auth.py"

tests:

- test_id: T-AU-001

name: "test_login_valid_credentials"
action: "POST /api/v1/auth/login with valid admin credentials"
expected: "200, response contains access_token and refresh_token"

- test_id: T-AU-002

name: "test_login_invalid_password"
action: "POST /api/v1/auth/login with wrong password"
expected: "401, error=INVALID_CREDENTIALS"

- test_id: T-AU-003

name: "test_protected_endpoint_without_token"
action: "GET /api/v1/works without Authorization header"
expected: "401"

- test_id: T-AU-004

name: "test_protected_endpoint_with_valid_token"
action: "GET /api/v1/works with valid JWT"
expected: "200"

- test_id: T-AU-005

name: "test_rbac_state_user_scope"
action: "Login as state_user (Maharashtra), GET /api/v1/constituencies"
expected: "Only Maharashtra constituencies returned"

- test_id: T-AU-006

name: "test_rbac_mp_cannot_access_other_constituency"
action: "Login as mp_user (Pune), GET /api/v1/constituency/Mumbai"
expected: "403"

- test_id: T-AU-007

name: "test_account_lockout_after_5_failures"
action: "5 failed login attempts for same user"
expected: "6th attempt returns 429"

- category: "API Integration Tests"

test_file: "tests/test_api.py"

tests:

- test_id: T-API-001

name: "test_works_list_pagination"
action: "GET /api/v1/works?page=1&per_page=10"
expected: "Returns exactly 10 records, pagination.total_pages correct"

- test_id: T-API-002

name: "test_works_filter_by_state"
action: "GET /api/v1/works?state=Maharashtra"
expected: "All returned works belong to Maharashtra constituencies"

- test_id: T-API-003

name: "test_works_sort_by_risk_score"
action: "GET /api/v1/works?sort_by=risk_score&sort_order=desc"
expected: "Records sorted by risk_score descending"

- test_id: T-API-004

name: "test_anomalies_list"
action: "GET /api/v1/anomalies"
expected: "Returns list of anomalies with correct schema"

- test_id: T-API-005

name: "test_anomaly_status_update"

action: "PATCH /api/v1/anomalies/{id} with {status: ACKNOWLEDGED}"

expected: "200, anomaly status updated, audit_log entry created"

- test_id: T-API-006

name: "test_analytics_national_summary"

action: "GET /api/v1/analytics/national-summary"

expected: "Returns total_works, total_expenditure, total_anomalies, risk_distribution"

- test_id: T-API-007

name: "test_report_generation"

action: "POST /api/v1/reports/generate with {scope: CONSTITUENCY, scope_id: X, format: CSV}"

expected: "Returns report_id and download_url; download returns valid CSV"

- test_id: T-API-008

name: "test_data_upload_endpoint"

action: "POST /api/v1/admin/upload with valid CSV file"

expected: "Returns upload_id, records_valid > 0"

- test_id: T-API-009

name: "test_synthetic_data_generation_endpoint"

action: "POST /api/v1/admin/generate-synthetic with {num_constituencies: 5, num_works_per_constituency: 20}"

expected: "200, database populated with 100 works"

- test_id: T-API-010

name: "test_health_endpoint"

action: "GET /health"

expected: "200, body contains database_connected=true"

7.2 Anomaly Detection Validation Metrics

YAML

ML_VALIDATION:

dataset: "Synthetic data with ground truth labels (from FR-DIM-002)"

metrics:

cost_overrun_detection:

precision_target: ">= 0.70"

recall_target: ">= 0.80"

f1_target: ">= 0.75"

evaluation: |

predicted = anomalies WHERE anomaly_type = 'COST_OVERRUN'

actual = synthetic_anomaly_labels WHERE anomaly_type = 'COST_OVERRUN' AND anomaly_injected = true

precision = TP / (TP + FP)

recall = TP / (TP + FN)

f1 = 2 * (precision * recall) / (precision + recall)

duplicate_detection:

precision_target: ">= 0.75"

recall_target: ">= 0.80"

evaluation: |

predicted = duplicate_pairs

actual = synthetic_anomaly_labels WHERE anomaly_type = 'DUPLICATE_WORK'

Match based on work_id involvement in pairs

delayed_project_detection:

precision_target: ">= 0.90"

recall_target: ">= 0.95"

note: "Higher targets because delay detection is rule-based with clear thresholds"

fund_utilization_detection:

precision_target: ">= 0.80"

recall_target: ">= 0.85"

overall_anomaly_detection:

precision_target: ">= 0.70"

recall_target: ">= 0.80"

note: "Across all anomaly types combined"

validation_script:

location: "scripts/validate_detection.py"

execution: "python scripts/validate_detection.py"

output: |

=====

MPLADS Sentinel - Anomaly Detection Validation

=====

Dataset: 5000 works, 400 injected anomalies

COST_OVERRUN:

Precision: 0.XX (target: ≥ 0.70) [PASS/FAIL]

Recall: 0.XX (target: ≥ 0.80) [PASS/FAIL]

F1: 0.XX (target: ≥ 0.75) [PASS/FAIL]

DUPLICATE_WORK:

Precision: 0.XX (target: ≥ 0.75) [PASS/FAIL]

Recall: 0.XX (target: ≥ 0.80) [PASS/FAIL]

DELAYED_PROJECT:

Precision: 0.XX (target: ≥ 0.90) [PASS/FAIL]

Recall: 0.XX (target: ≥ 0.95) [PASS/FAIL]

FUND_UTILIZATION:

Precision: 0.XX (target: ≥ 0.80) [PASS/FAIL]

Recall: 0.XX (target: ≥ 0.85) [PASS/FAIL]

OVERALL:

Precision: 0.XX (target: ≥ 0.70) [PASS/FAIL]

Recall: 0.XX (target: ≥ 0.80) [PASS/FAIL]

VALIDATION RESULT: [ALL PASS / X FAILURES]

=====

7.3 Frontend Validation Checklist

YAML

FRONTEND_VALIDATION:

accessibility:

tool: "axe-core (via @axe-core/react or browser extension)"

targets:

critical_violations: 0

serious_violations: 0

moderate_violations: " ≤ 5 "

pages_to_test:

- "/login"
- "/dashboard/national"
- "/dashboard/constituency/{id}"
- "/alerts"

responsiveness:

breakpoints_to_verify:

- width: 1920
label: "Full HD desktop"
expected: "All elements visible, map full width"
- width: 1024
label: "Small desktop/tablet landscape"
expected: "Layout adjusts, charts may stack"
- width: 768
label: "Tablet portrait"
expected: "Single column layout, all content accessible"
- width: 375
label: "Mobile"
expected: "Navigation collapses to hamburger, charts scrollable"

cross_browser:

browsers:

- "Chrome latest (primary)"
- "Firefox latest"

verification: "All pages render correctly, all interactions work"

functional_ui_checklist:

- check: "Login form submits and redirects to dashboard"
page: "/login"
- check: "KPI cards show non-zero values after data load"
page: "/dashboard/national"
- check: "Map renders with colored states"
page: "/dashboard/national"
- check: "Map click navigates to state view"
page: "/dashboard/national"
- check: "Alert table shows data with severity badges"
page: "/dashboard/national"

- check: "Alert severity filter works"
page: "/alerts"
- check: "Constituency detail page loads with works table"
page: "/dashboard/constituency/{id}"
- check: "Works table sorting works on risk_score column"
page: "/dashboard/constituency/{id}"
- check: "CSV export downloads file"
page: "/dashboard/constituency/{id}"
- check: "Charts render without errors (no blank spaces)"
page: "all dashboard pages"
- check: "Logout clears session and redirects to login"
page: "any page"

7.4 End-to-End Demo Validation Script

YAML

E2E_DEMO_SCRIPT:

title: "MPLADS Sentinel - Complete Demo Flow"

prerequisites:

- "Docker Compose running with all services healthy"
- "No pre-existing data (fresh deployment)"

steps:

- step: 1

action: "Navigate to https://localhost (or configured URL)"
expected: "Login page displayed"
verification: "Login form visible with username/password fields"
- step: 2

action: "Login with admin/Admin@1234"
expected: "Redirect to national dashboard"
verification: "Dashboard loads, KPI cards show 0 (no data yet)"
- step: 3

action: "Navigate to Admin panel → Generate Synthetic Data"
expected: "Synthetic data generator form"
action_detail: "Click 'Generate' with defaults (50 constituencies, 100 works each)"
expected_result: "Success message: 5000 works generated"
verification: "KPI 'Total Works Analyzed' now shows 5,000"

- step: 4

action: "Navigate to Admin panel → Run Anomaly Detection"

expected: "Pipeline starts executing"

action_detail: "Click 'Run Detection Pipeline'"

expected_result: "Success: Pipeline completed, X anomalies detected"

verification: "KPI 'Anomalies Detected' shows non-zero value"

- step: 5

action: "View national dashboard"

expected: |

- KPI cards populated with correct values
- India map shows colored states based on risk
- Top 10 risk constituencies bar chart visible
- Anomaly type distribution donut chart visible
- Alert feed shows recent alerts sorted by severity

verification: "All dashboard sections render with data"

- step: 6

action: "Click on a HIGH/CRITICAL state on the map"

expected: "Navigate to state view showing that state's constituencies"

verification: "Only constituencies from clicked state shown"

- step: 7

action: "Click on a high-risk constituency"

expected: "Constituency detail page loads"

verification: |

- KPIs specific to constituency
- Works table with risk scores
- Duplicate pairs displayed (if any)
- Risk breakdown radar chart

- step: 8

action: "In works table, sort by risk score descending"

expected: "Highest risk works appear first"

verification: "First work has higher risk_score than second"

- step: 9

action: "Click on a work with COST_OVERRUN anomaly"

expected: "Work detail shows cost overrun percentage, severity, detection method"

verification: "actual_expenditure > sanctioned_amount * 1.15"

- step: 10

action: "Navigate to Alerts page"

expected: "Full list of alerts with filters"

action_detail: "Filter by severity=CRITICAL"

expected_result: "Only CRITICAL alerts shown"

verification: "All visible alerts have CRITICAL severity badge"

- step: 11

action: "Click on a CRITICAL alert → Change status to ACKNOWLEDGED"

expected: "Status updates, audit trail shows entry"

verification: "Audit trail shows: OLD=NEW, NEW=ACKNOWLEDGED, user=admin, timestamp"

- step: 12

action: "Generate PDF report for a constituency"

expected: "Report generates and downloads"

verification: "PDF opens, contains executive summary, top risk works, charts"

- step: 13

action: "Generate CSV export from works table"

expected: "CSV downloads with all work data"

verification: "CSV has correct number of rows and columns"

- step: 14

action: "Logout"

expected: "Redirect to login page, previous session token invalid"

verification: "Attempt to access /dashboard → redirected to /login"

- step: 15

action: "Login as state_user (Maharashtra)"

expected: "Dashboard shows only Maharashtra data"

verification: "KPIs reflect Maharashtra-only data, map highlighted only for Maharashtra"

- step: 16

action: "As state_user, attempt to access another state's constituency"

expected: "403 Forbidden"

verification: "Error page or redirect indicating insufficient permissions"

overall_pass_criteria:

- "All 16 steps execute successfully"

- "No JavaScript console errors"

- "No server 500 errors in logs"

- "Page transitions take < 3 seconds"

- "All data displayed is consistent with database records"

7.5 Submission Readiness Checklist

YAML

SUBMISSION_CHECKLIST:

code_quality:

- item: "No linting errors (flake8 for Python, ESLint for JS)"

command: "flake8 . && cd frontend && npx eslint src/"

status: "PASS required"

- item: "All imports used, no dead code"

command: "Manual review + flake8 --select F401"

status: "PASS required"

- item: "Environment variables documented"

file: ".env.example"

status: "PASS required"

- item: "No hardcoded secrets in code"

command: "grep -r 'password\\|secret\\|key' --include='*.py' --include='*.js' --include='*.ts'"

status: "PASS required (only .env references)"

documentation:

- item: "README.md with project overview"

sections: ["Project Description", "Architecture", "Tech Stack", "Setup Instructions", "Demo Guide", "API Docs"]

status: "PASS required"

- item: "Docker Compose setup instructions"

content: "Step-by-step from git clone to running system"

status: "PASS required"

- item: "API documentation accessible"

url: "/api/v1/docs"

status: "PASS required"

deployment:

- item: "docker-compose up works from clean state"

command: "docker-compose down -v && docker-compose up --build -d"

verification: "All services healthy within 60 seconds"

status: "PASS required"

- item: "Database migrations run automatically on startup"

verification: "All tables created after fresh docker-compose up"

status: "PASS required"

- item: "Seed users created automatically"

verification: "Login with admin/Admin@1234 works on fresh deployment"

status: "PASS required"

testing:

- item: "Backend tests pass"

command: "docker-compose exec api pytest tests/ -v"

target: ">= 70% pass rate, 0 critical failures"

status: "PASS required"

- item: "Anomaly detection validation passes"

command: "docker-compose exec api python scripts/validate_detection.py"

target: "All precision/recall targets met"

status: "PASS required"

- item: "Accessibility audit passes"

tool: "axe-core"

target: "0 critical/serious violations"

status: "PASS required"

functional_completeness:

- item: "Data ingestion (CSV upload) works"
status: "MUST"
- item: "Synthetic data generation works"
status: "MUST"
- item: "Cost overrun detection works"
status: "MUST"
- item: "Duplicate work detection works"
status: "MUST"
- item: "Delayed project detection works"
status: "MUST"
- item: "Fund utilization anomaly detection works"
status: "MUST"
- item: "Risk scoring works"
status: "MUST"
- item: "National dashboard renders"
status: "MUST"
- item: "Constituency detail page renders"
status: "MUST"
- item: "Alert management works"
status: "MUST"
- item: "Authentication and RBAC works"
status: "MUST"
- item: "Report generation works"
status: "MUST"
- item: "Suspicious pattern detection works"
status: "SHOULD"
- item: "Web scraping works"
status: "SHOULD"

7.6 Success Metrics

YAML

SUCCESS_METRICS:

quantitative:

- metric: "Anomaly Detection Accuracy"
measurement: "F1 score on synthetic data with ground truth"
target: ">= 0.75 overall"
weight: "30%"

- metric: "System Response Time"

measurement: "95th percentile API latency"

target: "< 500ms"

weight: "15%"

- metric: "Dashboard Load Time"

measurement: "Time to Interactive"

target: "< 3 seconds"

weight: "10%"

- metric: "Feature Completeness"

measurement: "MUST_HAVE features implemented / total MUST_HAVE features"

target: ">= 90% (11/12 MUST_HAVE features)"

weight: "25%"

- metric: "Test Coverage"

measurement: "Backend line coverage"

target: ">= 70%"

weight: "10%"

- metric: "Accessibility Score"

measurement: "axe-core violations"

target: "0 critical, 0 serious"

weight: "10%"

qualitative:

- metric: "Innovation"

criteria: "Novel application of NLP for duplicate detection, multi-method anomaly detection ensemble"

- metric: "Relevance"

criteria: "Directly addresses MPLADS monitoring challenges with practical, deployable solution"

- metric: "Scalability Potential"

criteria: "Architecture supports 100x data growth with minimal changes"

- metric: "User Experience"

criteria: "Intuitive dashboard, clear risk communication, actionable alerts"

APPENDIX A: FILE STRUCTURE

text

mplads-sentinel/

├─ docker-compose.yml

├─ .env.example

├─ README.md

├─ docs/

| └─ SRS.md

| └─ architecture.md

| └─ demo-script.md

| └─ erd.png

├─ backend/

| └─ Dockerfile

| └─ requirements.txt

| └─ alembic/

| | └─ alembic.ini

| | └─ versions/

| └─ app/

| | └─ __init__.py

| | └─ main.py # FastAPI application entry

| | └─ config.py # Settings and environment

| | └─ database.py # Database connection

| | └─ models/ # SQLAlchemy models

| | | └─ __init__.py

| | | └─ work.py

| | | └─ constituency.py

| | | └─ fund_release.py

| | | └─ anomaly.py

| | | └─ user.py

| | | └─ audit_log.py

| | └─ schemas/ # Pydantic schemas

| | | └─ __init__.py

| | | └─ work.py

| | | └─ anomaly.py

```
| | | ├── auth.py
| | | └── analytics.py
| | ├── api/          # API routes
| | | ├── __init__.py
| | | ├── auth.py
| | | ├── works.py
| | | ├── constituencies.py
| | | ├── anomalies.py
| | | ├── analytics.py
| | | ├── admin.py
| | | └── reports.py
| | ├── services/     # Business logic
| | | ├── __init__.py
| | | ├── data_ingestion.py
| | | ├── synthetic_generator.py
| | | ├── anomaly_detection/
| | | | ├── __init__.py
| | | | ├── cost_overrun.py
| | | | ├── duplicate_detection.py
| | | | ├── delay_detection.py
| | | | ├── fund_utilization.py
| | | | ├── pattern_detection.py
| | | | └── pipeline.py
| | | ├── risk_scoring.py
| | | └── report_generator.py
| | ├── auth/         # Auth utilities
| | | ├── __init__.py
| | | ├── jwt_handler.py
| | | ├── rbac.py
| | | └── password.py
| | └── utils/
| | | ├── __init__.py
| | | └── audit.py
| ├── scripts/
| | ├── seed_users.py
| | └── validate_detection.py
```

```
| | └─ generate_synthetic.py
| └─ tests/
|   ├── __init__.py
|   ├── conftest.py
|   ├── test_data_validation.py
|   ├── test_anomaly_detection.py
|   ├── test_risk_scoring.py
|   ├── test_auth.py
|   └─ test_api.py
└─ frontend/
  ├── Dockerfile
  ├── package.json
  ├── vite.config.ts
  ├── tsconfig.json
  ├── public/
  | | └─ india-states.geojson
  | └─ └─ favicon.ico
  └─ src/
    ├── App.tsx
    ├── main.tsx
    ├── api/      # API client
    | | └─ client.ts
    | | └─ auth.ts
    | | └─ works.ts
    | | └─ anomalies.ts
    | | └─ └─ analytics.ts
    └─ components/
      ├── Layout/
      ├── Dashboard/
      | | └─ KPICards.tsx
      | | └─ IndiaMap.tsx
      | | └─ RiskBarChart.tsx
      | | └─ AnomalyDonut.tsx
      | | └─ TrendChart.tsx
      | | └─ └─ AlertFeed.tsx
      └─ Constituency/
```

```
| | | └─ ConstituencyDetail.tsx
| | | └─ WorksTable.tsx
| | | └─ RiskRadar.tsx
| | | └─ DuplicatePairs.tsx
| | └─ Alerts/
| | | └─ AlertList.tsx
| | | └─ AlertDetail.tsx
| | └─ Admin/
| | | └─ DataUpload.tsx
| | | └─ SyntheticGenerator.tsx
| | | └─ DetectionTrigger.tsx
| | └─ Auth/
| | | └─ LoginForm.tsx
| └─ contexts/
| | └─ AuthContext.tsx
| └─ hooks/
| | └─ useAuth.ts
| | └─ useApi.ts
| └─ pages/
| | └─ LoginPage.tsx
| | └─ NationalDashboard.tsx
| | └─ StateDashboard.tsx
| | └─ ConstituencyPage.tsx
| | └─ AlertsPage.tsx
| | └─ AdminPage.tsx
| └─ types/
| | └─ index.ts
| └─ utils/
| | └─ formatters.ts
| | └─ constants.ts
└─ nginx/
  └─ nginx.conf
```

APPENDIX B: DOCKER COMPOSE CONFIGURATION

YAML

docker-compose.yml

version: '3.8'

services:

postgres:

image: pgvector/pgvector:pg16

environment:

POSTGRES_DB: mplads_sentinel

POSTGRES_USER: mplads

POSTGRES_PASSWORD: \${DB_PASSWORD:-mplads_secure_password}

volumes:

- postgres_data:/var/lib/postgresql/data

ports:

- "5432:5432"

healthcheck:

test: ["CMD-SHELL", "pg_isready -U mplads -d mplads_sentinel"]

interval: 10s

timeout: 5s

retries: 5

redis:

image: redis:7-alpine

ports:

- "6379:6379"

healthcheck:

test: ["CMD", "redis-cli", "ping"]

interval: 10s

timeout: 5s

retries: 5

api:

build:

context: ./backend

dockerfile: Dockerfile

environment:

- DATABASE_URL=postgresql+asyncpg://mplads:\${DB_PASSWORD:-mplads_secure_password}@postgres:5432/mplads_sentinel

- REDIS_URL=redis://redis:6379/0

- JWT_SECRET_KEY=\${JWT_SECRET_KEY:-change_me_in_production}
- ENVIRONMENT=development
- LOG_LEVEL=INFO

ports:

- "8000:8000"

depends_on:

postgres:

- condition: service_healthy

redis:

- condition: service_healthy

command: >

```
sh -c "alembic upgrade head &&  
python scripts/seed_users.py &&  
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload"
```

celery_worker:

build:

context: ./backend

dockerfile: Dockerfile

environment:

- DATABASE_URL=postgresql+asyncpg://mplads:\${DB_PASSWORD:-mplads_secure_password}@postgres:5432/mplads_sentinel

- REDIS_URL=redis://redis:6379/0

depends_on:

postgres:

- condition: service_healthy

redis:

- condition: service_healthy

command: celery -A app.celery_app worker --loglevel=info --concurrency=2

frontend:

build:

context: ./frontend

dockerfile: Dockerfile

ports:

- "3000:80"

depends_on:

- api

nginx:

image: nginx:alpine

ports:

- "80:80"

volumes:

- ./nginx/nginx.conf:/etc/nginx/conf.d/default.conf

depends_on:

- api

- frontend

volumes:

postgres_data:

END OF SOFTWARE REQUIREMENTS SPECIFICATION

Document Status: COMPLETE — Ready for AI Agentic Development System Execution

Version: 1.0.0-MVP | Classification: Government Internal Use | Standard: IEEE 830 Adapted